A novel argument for cyclicity from the (in)visibility of infixes at morpheme junctures: Bottoms up![*]

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Accepted, *Natural Language and Linguistic Theory*

Abstract

This paper investigates what happens to grammatical relationships between two morphemes when an infix incidentally lands between them. Most relationships—including allomorphic and morphophonological relationships—persist, showing that these relationships are established before the intrusion of the infix. However, infixes do disrupt word-level phonology, which thus must be established after the intrusion of the infix. These findings furnish a novel argument for grammatical models that are (i) cyclic and bottom-up (or inside out) with respect to both exponence and certain (morpho)phonological operations, and (ii) noncyclic with respect to certain purely phonological processes, which are computed only at specific junctures (e.g., the word). A further argument from the interaction of infixation with object incorporation supports a post-syntactic model of morphology, such as Distributed Morphology. This work brings together earlier case studies of Hunzib (Kalin 2022b), Nancowry (Kalin 2023), and Palauan (Embick 2010) with several more languages, including a particularly extensive new study of the Bolivian language isolate Movima (drawing on Haude 2006, 2019), facilitating broader and stronger conclusions.

[*]Thank you to Byron Ahn, Nikita Bezrukov, Bronwyn Bjorkman, Jonathan Bobaljik, Hossep Dolatian, David Embick, Steven Foley, Florian Lionnet, Jack Merrill, Irina Monich, Neil Myler, Heather Newell, Nicholas Rolle, and Milena Šereikaitė for illuminating discussions of this (and closely-related) work, as well as the questions and suggestions of audiences at the following venues (chronologically listed): PSST 2021, UCLA's Syn/Sem, AIMM 5, NELS 52, ZAS's Syntax Circle, MSU, TAU, and UC Berkeley's Phorum. This paper has also benefitted significantly from the review process at NLLT, including the invaluable input of three anonymous reviewers and the Editor-in-Chief, Daniel Harbour.

1 Introduction

In this paper, I use infixes as a tool to investigate the timing of different types of local grammatical relationships and interactions. Locality plays a fundamental role throughout the grammar, constraining everything from syntactic selection to phonological assimilation. Importantly, *surface* (non)locality, that is, locality (or not) in the surface string, is not always an indication of the sort of locality that matters for a given relationship or interaction, as locality can hold at different derivational stages or representational levels. A classic example of this comes from the domain of idioms (e.g., Chafe 1970, Bach 1974, Chomsky 1995, O'Grady 1998), (1). The survival of the idiomatic meaning of *the shit hit the fan* in (1a) diagnoses a raising environment: *the shit* originates as the embedded subject, local to the rest of the idiom. The failure of the idiomatic interpretation in (1b) diagnoses a control environment: *the shit* originates as the matrix subject, not local to the rest of the idiom at any stage/level.

(1) a. [The shit]$_i$ continued t_i to hit the fan. (idiom survives)
b. #[The shit]$_i$ decided PRO$_i$ to hit the fan. (idiom does not survive)

More generally, the survival of a local relationship or interaction in a surface nonlocal configuration is a useful diagnostic tool, indicating that the relevant locality must hold at some earlier derivational point or distinct level of representation.

Infixes are affixes that, by their nature, disrupt (at least surface) locality. In canonical cases, they disrupt the linear integrity of another morph(eme), taking two parts of an exponent and pushing them apart. Consider the nominalizing infix *-ni-* in the Austronesian language Leti, (2) (Blevins 1999:386):[1]

(2) *-ni-* (NMLZ) + kakri (cry) → k<*ni*>akri 'act of crying'

In (2), the infix pushes apart the (independently meaningless) phonological substrings *k* and *akri* of the root *kakri*. The surface location of an infix in a string can be described with respect to a phonological or prosodic PIVOT in the infix's stem[2] (Ultan 1975, Moravcsik 2000, Yu 2007, i.a.). The pivot of the nominalizing infix *-ni-* in Leti is the first vowel of its stem (Blevins 1999).

For most linguists, the word "infix" conjures up cases like Leti, (2), where an infix is properly contained within (i.e., flanked by substrings of) a root. However, given the phonological/prosodic nature of infix placement, it should come as no surprise that an infix—when combining with a

[1]As is conventional, I enclose infixes in angled brackets in their surface locations, and when discussing an infix in isolation, I use dashes on either side of the exponent. Infixes of interest are italicized throughout the paper. I use Leipzig glossing conventions with the following additions: ABSL = absolute state, DIR = direct voice, G = G-stem former (Eton), INOM = instrumental nominalizer, INT = intensifier, INTENT = intentional, LINK = nasal linker, MID = middle voice, MLT = multiple event, PRC = process verbalizer, VLC = locative verbalizer, VM = verb marker, VPL = verbal plural.

[2]Throughout this paper, the term "stem" refers to the morphological form that a given affix attaches to, and is thus co-extensive with the term "base" in its common usage. This usage of the term "stem", in particular to pick out the morphological constituent an infix combines with, follows the literature on infixation. Of course, the term "stem" has other meanings in certain descriptive and theoretical traditions. In this paper, the only *other* usage of the term "stem" is in the fixed expression "stem-level", which refers to a particular morphological level in Stratal Optimality Theory.

complex (multimorphemic) stem—can appear within an affix or even at a juncture between morphemes in its stem, simply because its pivot happens to be found there. The former can be seen in Leti: when a root bearing the reciprocal prefix *va-* is nominalized, the nominalizing infix *-ni-* from (2) shows up inside this prefix (Blevins 1999:400):

(3) *-ni-* (NMLZ) + va-kini (RECP-kiss) → v<*ni*>a-kini 'reciprocal kissing'

The nominalizer *-ni-* appears before the first vowel of its stem, whether that vowel is found in a root, (2), or an affix, (3); the infix is totally oblivious to the morphological structure in its stem, caring only about the phonological string.

Infixes can also show up *between* morphemes. Consider the Palauan past tense infix, *-il-*, which surfaces after the first segment of its stem. In (4a), the infix combines with a simplex stem, a root, and appears inside that root. In (4b), the infix combines with a complex stem consisting of a prefixal "verb marker" and the root, and the infix appears between the prefix and the root (Flora 1974:74):[3]

(4) a. *-il-* (PST) + dasaʔ (carve) → d<*il*>asaʔ 'carved (participle)'
    b. *-il-* (PST) + m-dasaʔ (VM-carve) → m-<*il*>dasaʔ 'carved (middle voice)'

The intermorphemic location of *-il-* in (4b) is entirely incidental: *-il-* appears after the first segment of its stem (a phonologically defined location) without regard to morphological boundaries.

In this paper, I investigate issues of locality in conjunction with Palauan-like cases of infixation, where a language furnishes the right conditions for a canonical infix to (occasionally) appear between morphemes in its stem. The framing question of the investigation is stated in (5).

(5) When an infix surfaces (incidentally) between two morphemes in its stem, what happens to relations at/across that morpheme juncture that we would expect to be strictly local at some derivational stage or representational level?

Answering this question of infix (in)visibility has the potential to shed light on the derivational timing (or distinct representational levels) of different grammatical relationships and interactions.

The paper and main findings will unfold as follows. First, in §2, I provide an overview of the theoretical landscape and the predictions made by different theories with respect to the question in (5), drawing a crucial distinction among bottom-up, multistage, and once-and-for-all types of computations. Then in §3, I bring seven case studies of infixes (from six languages, five language families) to bear on these predictions, examining what happens to different types of relationships/interactions in the face of an intruding infix. Proceeding in §3 from the more syntactic/semantic side of things to the more phonological, the results are as follows:

(6) a. Semantic relationships *survive* (§3.1)
    b. Morphosyntactic relationships *survive* (§3.2)
    c. Allomorphic (suppletive) relationships *survive* (§3.3)

[3]Embick (2010:104–107) first brought the theoretical significance of this Palauan case study to light. I will return to it at length in §3.3.1.

d. Morphophonological interactions *survive* (§3.4)
e. Word-level phonological interactions *do not survive* (§3.5)

The core finding, summarizing (6), is that an infix is invisible for all relationships and interactions in its stem except word-level phonology.

Diving in to these findings a little more deeply, the investigation reveals that despite "canceled" adjacency (on the surface), a number of locality-dependent relationships/interactions persist when an infix intrudes between the interacting morphemes: (i) the two morphemes can still be semantically interpreted together (even as an idiom), (ii) they can still be in tight morphosyntactic relationships with each other (e.g., selection), (iii) one morpheme can still condition suppletive allomorphy of the other (even when the conditioning environment is phonological!), and (iv) the two morphemes can still interact phonologically, in particular, for phonological processes that appear to be morphologically conditioned. The survival of all these relationships/interactions shows that they must hold at some derivational point (or distinct level of representation) that precedes (or otherwise excludes) the intrusion of the infix. However, the "canceled" adjacency of two morphemes *does* affect some kinds of locality-dependent interactions: word-level phonology in the stem of an infix *is* disrupted by the intrusion of the infix. This shows that these interactions care about surface locality, that is, they must be established at some derivational point (or level of representation) that follows (or otherwise includes) the infix.

In §4.1, I follow up on the predictions of different theories from §2, in light of the findings presented in §3. Some of the findings about (non)disruptability of relationships/interactions are predicted by all theories, like the imperviousness of semantic interpretation to the intrusion of an infix. (Indeed, it would be very disturbing to discover an infix that disrupts the semantics of its stem!) Others of the findings, such as the imperviousness of phonologically conditioned suppletive allomorphy to the intrusion of an infix, are very surprising from the perspective of certain influential theories. For example, this is surprising if morphological and/or phonological operations are computed only once (noncyclically), as in classic parallel Optimality Theory (OT). In a nutshell, the findings constitute a strong novel argument that exponence and some phonological processes are computed cyclically, from the bottom of a morphosyntactic structure upward (or from the inside of a word out), one morpheme/step at a time; on the other hand, at least some other phonological processes must not be computed cyclically, but rather once-and-for-all over a larger constituent.

Finally, §4.2 puts forward an argument from one case study, based on syntactic object incorporation feeding infixation, that exponence and (morpho)phonology are computed *after* the syntax. The findings, taken all together, are straightforwardly captured in a post-syntactic, realizational, derivational framework like Distributed Morphology, with the fine details in terms of the timing of exponence, infixation, and (morpho)phonology along the lines I have proposed in Kalin 2022a.

Before turning to the theoretical landscape and investigation, I use the remainder of the introduction to lay out the backdrop of the paper: a definition of infixation (§1.1) and a discussion of the language sample (§1.2).

1.1 What's an infix?

As a starting point for what it means to be an infix, I adapt the formulation of Blevins 2014:

(7) "[An infix is] a bound [exponent] whose phonological form consists minimally of a single segment [and which] is preceded and followed in at least some word-types by non-null segmental strings which together constitute a relevant form-meaning correspondence of their own, despite their nonsequential phonological realization."

This says that an infix is a morph that (i) is minimally the size of a segment (i.e., nonfloating), (ii) is bound (i.e., affixal), and (iii) can interrupt another morphological constituent, be that a root, as in (2) and (4a), or an internally complex form, as in (3) and (4b).

For my purposes in this paper, to be sure I'm identifying truly uncontroversial cases of infixation, I take two additional aspects of infixes to be definitional, (8):

(8) Further definitional properties of infixes for the present study
   a. An infix can appear inside a root (though need not always do so).
   b. An infix has a placement that can be described in purely phonological or prosodic terms.

According to Blevins' definition in (7), an infix need not ever appear inside a root to be considered an infix. While I believe Blevins is ultimately correct on this point, (8a) takes an extra step to ensure we're dealing with one and the same type of element, when comparing infixes across languages in this study. The additional component in (8b) incorporates a core aspect of infixes that is absent from the definition in (7), namely, that the location of an infix in its stem can be described as preceding or following a particular phonological or prosodic pivot, for example, C, V, or syllable (Ultan 1975, Moravcsik 2000, Yu 2007, i.a.). This condition aims to ensure that all cases of infixation included in this study have the same derivational timing/representational status, removing a potential confound in interpreting the results to do with displacement that's not entirely phonological in nature.

The focus in this paper is on a very particular subcase of infixation, namely, cases where an infix incidentally lands *between* morphemes in its stem. Importantly, I am not talking about hypothetical infixes that *seek out* an intermorphemic position (indeed this would violate both the criteria in (8)); I am also not talking about infixes that incidentally *always* have an intermorphemic position (which would violate the criterion in (8a)). Rather, the infixes I am investigating are run-of-the-mill infixes, conforming to all the criteria laid out above in (7) and (8). What is special about the infixation cases at hand here is just that a number of happy coincidences within a language enable an infix to combine with a complex stem, and when doing so, to sometimes land incidentally at a morpheme juncture. Palauan in (4) is like this, and we will of course see many more examples.

1.2 The infixes included in this study

I have found a total of seven morphs that (i) are infixes according to the definition in (7), (ii) have classic infixal distributions (phonologically or prosodically defined; can appear inside roots), as per the additional criteria in (8), and (iii) *also* can appear between morphemes in their stems under the right morphological/phonological circumstances. The seven corresponding abstract morphemes (which have these infixal realizations) are listed in Table 1 by language and family.

Table 1: Language sample

Family	Language	Morpheme(s)	Reference(s)
Austroasiatic	Katu	NOM	Costello 1991, 1998
	Nancowry	CAUS, INOM	Radhakrishnan 1981, Kalin 2023
Austronesian	Palauan	PST	Josephs 1975, Embick 2010
Movima (isolate)	Movima	IRR	Haude 2006
Niger-Congo	Eton	G-FORM	Van de Velde 2008
Northeast Caucasian	Hunzib	VPL	van den Berg 1995, Kalin 2022b

This is, of course, rather sparse for a typological sample. Infixes are scarce to begin with (compared to prefixes and suffixes), and the type of infix relevant here is significantly rarer than that. The list in Table 1 exhaustively reflects extensive engagement with typological studies (most notably Yu 2007) and many grammars and articles over several years. In the sample's favor are its geographic and genetic diversity, with four of the six languages on different continents as well as in different language families. The only two languages that share a family—Katu and Nancowry—also share a continent, but nevertheless belong to different branches of that family and are nearly a thousand miles (of largely ocean) apart. There is diversity, too, in the grammatical functions of the affixes, ranging from low derivational morphology to high inflectional morphology. I share the sentiment articulated in Harbour 2016, "I take patterns that favor my theory seriously only when they are robust in the sense of occurring across a geographical, genetic, and grammatical range" (p. 6); I believe it is justified to take the present sample seriously, despite its sparsity.

As reflected in the references listed for each language, this work brings together earlier case studies of infixation in Hunzib (Kalin 2022b), Nancowry (Kalin 2023), and Palauan (Embick 2010:§3.4.3) with infixes in several more languages. The most prominent new case study in this paper is of the Bolivian language isolate Movima, for which I draw on Haude 2006. In bringing old and new case studies together, I cover more empirical ground, including bringing in more types of grammatical relations and more examples of specific grammatical relations. The findings synthesize and extend those of the studies I build on, enabling me to make broader and stronger claims about the architecture of the grammar.

For each infix in Table 1, given the documentation available, I have attempted to exhaustively investigate the different sorts of grammatical relationships and interactions that this infix might interrupt when it appears intermorphemically. These relationships/interactions include semantic relationships, morphosyntactic relationships, allomorphic (suppletive) relationships, and different kinds of phonological interactions. Not all of the infixes furnish examples of intrusion in every kind of relationship/interaction (at least not as far as I could discover), but all are relevant for more than one kind of relationship/interaction.

1.2.1 Excluded (but still relevant) case studies

There are three further potentially relevant case studies that I have excluded from the sample, on the basis of their not fitting all the definitional criteria laid out in §1.1, (7) and (8). For all three cases, I do *not* think the item in question should be labeled an "infix". I briefly discuss the case studies here as they will come up at a few junctures in the paper. Importantly, I am not excluding these three case studies because they go against the findings in the paper—quite to the contrary, they actually all fall in line with canonical infixes, thereby strengthening my findings.

The first excluded case study is the past tense morpheme in Turoyo (Jastrow 1993, Kalin 2020). Past tense marking in Turoyo is realized in a variable position that can be described mainly phonologically; it also shows up between morphemes, making it a potential candidate for inclusion. But, as I note (Kalin 2020:165–166), the positioning of the tense marker is sensitive to morphological boundaries as well as phonological factors, and indeed its distribution is not statable purely phonologically or prosodically. Relatedly, this tense marker never surfaces inside a root. Past tense marking in Turoyo is therefore not a canonical infix according to the strict inclusion criteria I am using in (8).

The second excluded case study is the well-known case of Cibemba applicative causatives (Hyman 1994), where the applicative morpheme intrudes (counter-scopally) between the causative morpheme and the root. This case study is famous for displaying cyclic phonological effects, since the intruding applicative morpheme surprisingly does *not* disrupt a particular (morpho)phonological interaction between the causative and the root. Although this has been analyzed by some as involving infixation (e.g. Orgun 1996, Yu 2007), the applicative never appears within a root, nor does it have a phonological/prosodic pivot that is driving its intrusion, again counter to (8). The applicative in Cibemba is not a canonical infix.

The final excluded case study is that of so-called expletive infixation in English, for example, as in *Massa<fucking>chusetts* and *abso<damn>lutely*. Simplifying somewhat, these expletives have a prosodic placement, preceding the foot bearing main stress in their stem (for discussion, see e.g. Siegel 1974, Aronoff 1976, McCawley 1978, McCarthy 1982). And, these expletives can indeed appear intermorphemically, for example, *in<fucking>edible*. The reason for their exclusion here is that the expletives that can be "infixed" are not affixes (they are free, not bound) and are often morphologically complex, counter to the criteria in (7). In addition, morpheme boundaries can affect their placement (as in *un<fucking>believable*), counter to (8), and—as is unusual for infixes—for some speakers, the expletive must both follow *and* precede a stressed syllable. Infixed expletives therefore do not fit the definitional aspects of being canonical infixes.

As mentioned above, these case studies conform to the larger findings about locality that are supported by the seven case studies of canonical infixes from Table 1. This suggests that the criteria I have laid out for identifying infixes in §1.1 are perhaps too stringent, or, that infixes belong to a larger morphological phenomenon that can subsume all of these cases. I leave this as a question for future work.

2 Theoretical backdrop: Predictions of different theories

Recall the question guiding this paper, repeated below from (5):

(9) When an infix surfaces (incidentally) between two morphemes in its stem, what happens to relations at/across that morpheme juncture that we would expect to be strictly local at some derivational stage or representational level?

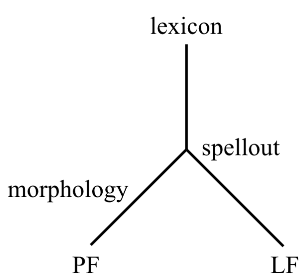
In this section, I explore where it is that theories come apart from each other, and where theories converge, with respect to predictions for different types of grammatical relationships/interactions that may take place inside the morphological form an infix combines with (its stem). I begin briefly with the syntactic and semantic side of the picture, §2.1, then more extensively turn to the morphological and phonological side, §2.2. This discussion will anchor the empirical survey in §3 and introduce some crucial terminology.

2.1 On the interaction of infixes with semantic and (morpho)syntactic relationships

All theories (to my knowledge) are unified in predicting that infixes should *not* disrupt relationships that are semantic or (morpho)syntactic in nature. In other words, semantic and (morpho)syntactic relationships that hold for two morphemes in the stem of an infix should survive intrusion of the infix between the morphemes.

Consider the mainstream generative Y model (originating in Chomsky and Lasnik 1977), with a post-syntactic morphology (as in Distributed Morphology):

(10)



In this model, the output of syntax—at the point of spellout—interfaces on the one hand with phonology and at least some aspects of morphology (the branch to PF), and on the other hand with semantics (the branch to LF). Since infixation is part of morphology/phonology, infixation should not affect anything that leads to the syntactic representation at spellout, and because of that, also not anything on the LF branch. This model therefore predicts no disruption of syntactic and semantic relationships by an infix.

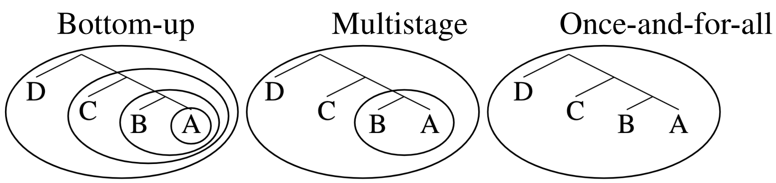
Though encoded in different ways, other models of the grammar—those that don't assume (10)—ultimately make the same prediction (e.g., Aronoff 1976, Di Sciullo and Williams 1987, Jackendoff 1997, Stump 2001, Starke 2009), that an infix should not disrupt syntactic and semantic relationships in its stem. Indeed, across a very diverse set of theoretical frameworks, the operation of infixation is sometimes invoked to explain the surprising appearance of some exponent in a position where it is not semantically interpreted and/or where it does not interrupt the (morpho)syntax (e.g., Rice 1987, Orgun 1996, Crysmann 2000), even when the exponent itself is not (at least obviously) an infix.

In the empirical survey in §3, I include semantic relationships (§3.1) and (morpho)syntactic relationships (§3.2) as a reality check and a proof-of-concept of the methodology, but the expected finding according to all theories is *lack* of disruption, and thankfully, this is borne out.

2.2 On the interaction of infixes with morphological and phonological relationships

On the morphology/phonology side of the picture, theories differ from each other along one crucial dimension that will be relevant here: (non)cyclicity. A cyclic computation is one that applies more than once, over increasingly sized domains. I make a three-way cyclicity distinction in terms of how a particular (type of) operation might be treated in a given theory, represented schematically in (11); the circles each delineate a derivational stage or representational level at which a particular computation (or type of computation) applies.

(11)



Bottom-up operations have many more opportunities for applying than once-and-for-all or even multistage operations. It is for this reason that (non)cyclicity is relevant for predicting whether the outcome of a given operation should, or should not, survive the intrusion of an infix.

At one extreme, a theory might be BOTTOM-UP (or inside out[4]) with respect to a particular type of operation. A bottom-up operation starts from the most deeply embedded component of a structure and works its way upwards, (re)applying with the addition of every node. Such a computation is in some sense *hyper*-cyclic, since it applies at every opportunity. In the bottom-up schematic in (11), the operation applies four times, once over just A, then A+B, then A+B+C, then over the whole structure, A+B+C+D. This is instantiated by the Distributed Morphology operation of Vocabulary Insertion, which involves inserting the best fit exponent into a terminal node, and which (following Bobaljik 2000) is commonly assumed to work bottom-up from the most embedded node, one node at a time. (See §4.1.1, (71), for a calculus for determining the "bottom" of a structure.) Bottom-up operations should, in general, survive the intrusion of an infix. In §3.3–4, I show that both exponence and morphologically-conditioned phonological processes

[4]I use the term "inside out" as the sort of equivalent of "bottom up" in frameworks where words don't have hierarchical morphological structures, but nevertheless words have an innermost stem/root and some operations apply from the stem/root moving outward, to the edges of the word. This is the case, for example, in Paradigm Function Morphology for how Realization Rules work. Henceforth, when I use the term bottom-up, it should be assumed that this is inclusive of inside-out theories unless explicitly stated otherwise.

always survive the intrusion of an infix, consistent with bottom-up computation of these operations.

At the opposite extreme, a theory might compute a particular type of operation ONCE-AND-FOR-ALL, that is, only one time, noncyclically. In the schematic in (11), the operation applies only over A+B+C+D. Most relevant in this paper will be once-and-for-all treatments of some computation(s) at the word level. This is how phonology operates in classic parallel OT (McCarthy and Prince 1993a,b), where phonological constraints evaluate an entire word, and not any sub-constituents of the word. Once-and-for-all operations should, in general, *not* survive the intrusion of an infix.[5] In §3.5, I show that word-level phonological processes do not survive the intrusion of an infix, consistent with once-and-for-all computation of these operations.

In between these extremes, there is a (vast swath of) middle ground: theories can treat an operation as applying more than once, cyclically, but not at every possible juncture. I will refer to this as a MULTISTAGE approach, as most commonly a particular computation (re-)applies at pre-defined stages/levels. In (11), the multistage schematic has a stage A+B and then a stage A+B+C+D, so the computation applies twice. Stratal OT (e.g., Kiparsky 2000, Bermúdez-Otero 2012, 2017) is a multistage theory of phonology, with multiple specific junctures—termed strata—at which phonological computations (re)apply, most relevant here being the stem-level and the word-level; there may also be further cycles of the stem-level computation, triggered by the addition of particular ("cyclic") stem-level morphemes. It's important to note that the "multistage" label encompasses a large and diverse set of theories, over which it is harder to generalize. But, in general, multistage operations should show mixed behavior, on the basis of whether the intruding infix is inside or outside a relevant stage/level.[6] No mixed behavior is observed in the empirical sample in §3.

To connect the above discussion to actual theories, the remainder of this section separately takes up the theoretical landscape of (i) exponence, §2.2.1, (ii) (morpho)phonological processes, §2.2.2, and (iii) infixation, §2.2.3, as these (types of) operations are all logically separate from each other. (Readers hoping to get to the meat of the data faster can safely skip directly to §3.)

2.2.1 The (non)cyclicity of exponence: theoretical landscape

EXPONENCE refers to the choice among, and insertion of, phonological exponents. For example, given the noun *dog*, plural is exponed as *-z*, but given the noun *child*, plural is exponed as *-ren*.[7]

[5] There is an exception to this prediction: it is possible (though counter to all actual existing theories, to my knowledge) to imagine a once-and-for-all approach that uniquely delays the operation of infixation relative to all other operations. If this is the case, then the outcome of those other operations would survive the intrusion of an infix.

[6] There are some approaches that don't neatly fit into any of the categories described above, but which would—like multistage approaches—predict mixed behavior of intrusion/nonintrusion of an infix, based on the particular local ordering of operations. For example, the Optimal Interleaving approach of Wolf 2008, a development of OT related to Harmonic Serialism (McCarthy 2000), treats exponence as applying bottom-up in the usual case, but allows for the possibility of exceptional simultaneous or top-down exponence as well (see also Deal and Wolf 2017). Bottom-up exponence should not be disrupted by the intrusion of an infix, but simultaneous or top-down exponence could be.

[7] I intend for the terms "expone" and "exponence" to be able to cover both the choice/insertion of exponents in a realizational theory—where abstract features/roots are separate from and in some sense prior to phonological exponents—as well as in an incremental theory—where abstract features/roots do not exist independently of their phonological exponents, but rather features/roots and exponents are inserted together. See Stump 2001:Ch. 1 on the incremental-realizational distinction.

When a given abstract element corresponds to more than one exponent, as with *-z* and *-ren*, this situation is referred to as SUPPLETIVE ALLOMORPHY, a phenomenon that revolves around choosing the right exponent in a particular context.

With respect to the operation of exponence, all three types of treatments schematized in (11) are attested in the literature, but bottom-up exponence is the most common. Bottom-up exponence is found across a diverse array of morphological frameworks, including Lexical Phonology and Morphology, Paradigm Function Morphology, Distributed Morphology, and Nanosyntax. (On these and more, see e.g. Anderson 1982, Kiparsky 1982, Lieber 1992, Wunderlich 1996, Bobaljik 2000, Stump 2001, Starke 2009, Embick 2010, Caballero and Inkelas 2013, Myler 2017, Choi and Harley 2019, Müller 2021, Kalin 2022a.) According to these theories, exponence proceeds hypercyclically, starting from the most deeply embedded node/component of a structure and working its way upwards/outwards, one exponent at a time.

There are also influential treatments of exponence that are once-and-for-all or multistage. In such theories, exponents within a particular domain are chosen simultaneously; this may apply to all exponents, just to exponents with a phonologically determined distribution, or (yet more restricted) just to exponents with a phonologically determined distribution that is optimizing. Once-and-for-all exponence is found most commonly in parallelist OT-based frameworks, where exponent choice is governed by the phonological grammar (e.g., Prince and Smolensky 1993, Mester 1994, Kager 1996, Mascaró 1996, 2007, Bonet et al. 2007, Smith 2015), as well as in other constraint-based approaches (e.g., Bonami and Crysmann 2016, Crysmann and Bonami 2016). Multistage exponence is also found mainly in OT-employing frameworks (e.g., Bermúdez-Otero 2012, Bye and Svenonius 2012, Svenonius 2012, 2016, Rolle 2020), where again a constraint-based grammar (usually the phonological grammar) is able to choose among competing exponents for a given morpheme, and this happens at specific pre-defined stages (e.g., strata, phases).

2.2.2 The (non)cyclicity of (morpho)phonological processes: theoretical landscape

(MORPHO)PHONOLOGY refers to phonologically derived effects that both (i) are not themselves meaning- or function-bearing, that is, these phonological alternations aren't exponents, and (ii) are triggered by phonological context and/or (at least apparently) morphological context. One basic way to characterize phonological alternations is based on this latter property: is the conditioning environment purely phonological, or is morphology implicated as well? An example of a purely phonological process is flapping in English, where alveolar stops become flaps in certain phonologically definable pre-vocalic environments. An example of a morphophonological process is the voicing of the final fricative in *leaf* (and a handful of other words) in the context of the plural suffix—*leaves*—which can at least descriptively be considered morphologically conditioned phonology.[8]

Overall the landscape for (morpho)phonological processes is very similar to that discussed above for exponence, though in this case the most prominent approaches are either once-and-for-all or multistage, and make use of an OT grammar. Exclusive once-and-for-all (morpho)phonological

[8] I am hedging here because there are many different ways to treat apparent morphophonology, some of which reduce it to pure phonology, and others of which reduce it to pure morphology. See more extensive discussion in §3.4.

computation is found in parallelist OT frameworks, where the whole word is considered together for the application of *all* word-internal (morpho)phonology (e.g., McCarthy and Prince 1993a,b, Benua 1997, Steriade 2008, Zukoff 2023). This is also how a sub-class of phonological processes is dealt with in certain theories. For example, in Kiparsky's Lexical Phonology and Morphology (Kiparsky 1982), the "post-lexical" phonology is noncyclic and applies over the whole word. ("Lexical" phonology, on the other hand, applies bottom-up.)

Multistage approaches to (morpho)phonology posit levels inside the word at which phonological computation happens. These levels may be determined by morphologically defined strata—for example, stem-level versus word-level—as in Stratal OT (e.g., Kiparsky 2000, Bermúdez-Otero 2012), or by some notion of (syntactic) phases (e.g., Bye and Svenonius 2012, Rolle 2020, Sande et al. 2020, Müller 2021). Such approaches are generally considered cyclic even though the OT grammar may involve a constraint re-ranking across levels, and so it is not usually an *identical* computation taking place at every level. Accordingly, in a multistage theory, a particular phonological alternation may be characterized as stem-level (cyclic) or word-level (noncyclic, in a sense), depending on whether it seems to apply word-internally or only at the word level.

On the other hand, bottom-up (morpho)phonological computation is found in a diverse set of frameworks, with at least some (morpho)phonological processes applying at every possible juncture, hyper-cyclically (e.g., Chomsky and Halle 1968, Kiparsky 1982, Booij and Rubach 1987, Stump 2001, Inkelas 2008, Kalin 2022a). These approaches often additionally posit a distinction between (morpho)phonological processes that are bottom-up and those that are multistage or once-and-for-all, with the latter types of (morpho)phonological processes *not* re-applying hyper-cyclically, but rather applying only at specific junctures (as described above).

How do exponence (§2.2.1) and (morpho)phonology relate to each other? Theories adopting exclusively once-and-for-all or multistage (morpho)phonology most often also make use of (at least some) once-and-for-all or multistage exponence, respectively, though this is not a logical necessity. On the other hand, theories positing at least some bottom-up (morpho)phonology most often make use of bottom-up exponence, with exponence interleaved with (morpho)phonology.

2.2.3 The timing of infixation itself: theoretical landscape

Infixation—the displacement of an infixal exponent into its stem—is of course a specific operation of interest in this paper. No theory (to my knowledge) takes infixation to have its own particular timing in the grammar, independent of other operations. Rather, infixation is always either timed (i) as applying alongside exponence of the infix, or (ii) as applying with the first application of (morpho)phonology after exponence of the infix. (It might also be that these two timing possibilities are the same, namely, if exponence is part of the phonological grammar.) Infixation thus inherits its (non)cyclicity from other operations.

As noted at the end of §2.2.2, most theories take exponence and (at least some) (morpho)phonological processes to pattern together in terms of their (non)cyclicity, making it straightforward to determine the (non)cyclicity of infixation as well. For example, the highly influential Prosodic Morphology approach to infixation features once-and-for-all exponence and (morpho)phonology, and includes infixation in this once-and-for-all computation (e.g., Prince and Smolensky 1993, McCarthy and Prince 1993a,b, Zoll 1996, Hyman and Inkelas 1997, Zukoff 2023). Conversely, in my own recent work (Kalin 2022a), I argue that exponence and some (morpho)phonology are computed from the bottom up, and that infixation happens during insertion of the infixal exponent, that is, infixation is also bottom-up. Throughout the paper, I will assume that infixation is in line with the timing of exponence/(morpho)phonology in a theory. I will, however, briefly consider a disconnect between infixation and other operations in §4.1.2, to see what the empirical predictions would be of pairing a once-and-for-all approach to exponence and (morpho)phonology with delayed infixation. (Spoiler: such a model will not be successful.)

There are a number of other, orthogonal debates in the literature on infixation, in terms of how this operation should be treated. One such debate revolves around the use of subcategorization to determine infix placement, and more broadly, about the role of the phonological grammar in infix placement. Another debate revolves around whether or not infixes are linearized as prefixes/suffixes prior to infixation. For extensive discussion of these debates, see Yu 2007:Ch. 2 and Kalin 2022a:§7. We will see in the end that the results of this paper do not have much to tell us in terms of these debates about infixation.

3 When does(n't) an infix disrupt local relationships and interactions?

The previous sections established a guiding question—about an infix's (in)visibility for various processes in its stem (the constituent it combines with), repeated in (12)—and a set of predictions.

(12) When an infix surfaces (incidentally) between two morphemes in its stem, what happens to relations at/across that morpheme juncture that we would expect to be strictly local at some derivational stage or representational level?

All theories agree that semantic and (morpho)syntactic relationships in the stem of an infix should not be disrupted by the intrusion of the infix (§2.1). Things are more complex in the domain of morphology and phonology: depending on the posited degree of cyclicity of an operation, theories make different predictions about whether an infix should or should not intrude in a (potential) instance of that operation applying in its stem (§2.2). (13) summarizes the crucial predictions.

(13) Predictions for morphological and phonological operations

  a. The intrusion of an infix between two morphemes in its stem should not affect cross-morpheme relations established by (i) *bottom-up* computation or (ii) *multistage* computation taking place at an earlier stage/level than the introduction of the infix.
  b. The intrusion of an infix between two morphemes in its stem should disrupt cross-morpheme relations established by (i) *once-and-for-all* computation or (ii) *multistage* computation taking place at a later stage/level than the introduction of the infix.

In this section, I set out to answer the question in (12), to see which of the predictions in (13) are borne out for different types of relationships and interactions. I begin with the proof-of-concept cases, with semantic relationships in §3.1 and (morpho)syntactic relationships in §3.2. The expected (§2.1) and borne-out finding for semantic and (morpho)syntactic relationships is that these relationships are *not* disrupted by the infixes at hand, which helps to validate the methodology as well as establish these infixes as true, canonical infixes. I then turn to the more theoretically informative cases, involving exponence, §3.3, morphophonology, §3.4, and other kinds of phonology, §3.5. In each section, I give a few representative examples from my case studies to illustrate the findings, and I consider the immediate implications in terms of which predictions are borne out. Throughout these sections, particular infixes are introduced in detail as they come up.

The subsection headings anticipate the overall findings, which—putting it all together—are that most types of local relationships/interactions survive infixation, that is, persist across an intruding infix, in line with bottom-up exponence and (morpho)phonology. Word-level phonology is the exception, *not* persisting across an intruding infix, in line with once-and-for-all phonological computation for at least some phonological processes. Multistage computation predicts mixed results of a sort which are not borne out in the case studies.

3.1 Semantic relationships (in the stem of an infix) survive infixation

Semantic interpretation—both compositional and idiomatic—is well known to be sensitive to interruption. Consider, for example, the contrast between (14a) and (14b).

(14) a. un-re-lock
b. re-un-lock

The relative closeness of *re-* and *un-* to the root determines which semantically composes with the root first—in (14a), *re-* has the closer relationship with the root, and so the meaning 'relock' is part of the overall meaning of the word (*un-* + 'relock'); but, in (14b) it is *un-* that is closer to the root, and so the meaning 'relock' is not a part of the meaning of the whole word (the meaning is instead *re-* + 'unlock').

Along similar lines, intervention in a word-level idiom destroys idiomaticity:

(15) Idiom: unearth (≈ 'discover')
a. re-un-earth (idiomatic reading possible)
b. #un-re-earth (idiomatic reading impossible)

The word in (15a) can have the meaning 'discover again', with the 'discover' meaning facilitated by the adjacency of *un-* and *earth*. In (15b), the 'discover' reading is impossible, corresponding to the *non*adjacency of *un-* and *earth* in this word. (Instead, this word can only mean something like 'take something out of the earth after having put it back in the earth'.) In the usual type of case, then, exemplified by (14) and (15), interpretation is sensitive to constituency and thus is highly interruptible.

What happens to semantic composition when the element intruding within a constituent is an infix? Recall from §2.1 that the expectation is that an infix should *not* interrupt semantic composition in its stem. The case study that I will focus on in this section is the irrealis morpheme in Movima, which I take up in §3.1.1. As discussed in §3.1.2, in *all* the case studies the core finding is the same: semantic interpretation of the stem of an infix is *not* interrupted by the infix, whether that interpretation is compositional or idiomatic.

3.1.1 Movima: Introduction and semantic relationships

Movima, a language isolate of Bolivia, is a predicate-initial and highly synthetic language with a rich variety of (mainly derivational) morphology (Haude 2006; henceforth H). The irrealis morpheme (H:§3.6.2, §10.3) is realized infixally and has a prosodically defined position that frequently lines up (incidentally) with a morpheme juncture. The irrealis morpheme will feature centrally throughout the paper, and so I will lay out its core properties in detail before turning back to the question at hand about semantic interpretation.

The irrealis morpheme in Movima is categorized by H as a mood marker, and accordingly it combines only with predicates and occupies an outer position in the structure of the word. The irrealis morpheme is, in fact, "the final word-forming process", with no affixes able to combine after it (H:81). The irrealis morpheme itself can combine with a verbal predicate, (16a-b), or a nominal predicate, (16c), and conveys irrealis mood or existential negation. Note that with predicate nominals the irrealis is *always* interpreted as existential negation, like in (16c), regardless of the overt presence of the negative morpheme *kas* (H:112,439–440).[9]

(16) a. sal<*a'*>mo ∅
<IRR>return 1SG
'I'll be back.' (H:438)

b. kas loy dej<*a'*>-na =∅
NEG INTENT <IRR>cook-DIR =1SG
'I won't cook anything.' (H:439)

c. (kas) en<*a'*>ferme:ra
NEG <IRR>nurse
'There is no nurse.' (H:80)

The irrealis morpheme is realized as an infix. The examples above show it can appear inside a root, (16a,c), or between morphemes, (16b).

In terms of its placement, the irrealis infix appears after the first foot of its stem, a foot which may consist of a single heavy syllable, (17a) (as well as all the examples in (16)), two light syllables, (17b), or a light syllable followed by a heavy syllable, (17c); I use a period to indicate the boundary after the first foot in the stems in (17).[10]

(17) a. -*(k)a'*- (IRR) + sal.mo (return) → sal<*a'*>mo (H:438)
b. -*(k)a'*- (IRR) + aro.so (rice) → aro<*ka'*>so (H:80)
c. -*(k)a'*- (IRR) + tivij.-ni (pain-PRC) → tivij<*a'*>-ni (H:79)

[9] When glossing infixes, in order to make it clear that the infix is displaced, I put the infix's gloss at the edge of the stem it is combining with—at the left edge of its stem if the infix's position is defined from the beginning of the stem (e.g., after the first foot), and at the right edge of its stem if the infix's position is defined from the end of the stem (e.g., after the final consonant). Note that my gloss-placement convention means that in cases of an intermorphemic infix, like in (16b), the gloss is not found at the juncture where the infix is, but rather still at the stem edge.

[10] As noted by H:78, "only the foot structure of the underived word matters." Placement of the infix is in fact often opaque, as the stem undergoes footing and (in the case of predicate nominals) penultimate lengthening prior to infixation, both of which may be obscured after post-infixation processes of (re-)footing and vowel shortening. The implications will be briefly discussed in fn. 45.

Post-consonantally, the infix has the form *-a'-*, as in (17a,c), and post-vocalically, it has the form *-ka'-*, as in (17b). The two forms of the infix, *-ka'-* and *-a'-*, arguably share a single underlying form: the initial consonant appears whenever otherwise there would be vowel hiatus. The variation between *-ka'-* and *-a'-* could be analyzed either as due to a morphologically conditioned phonological rule that inserts /k/ intervocalically when the irrealis infix creates vowel hiatus, or as due to the underlying representation of the exponent having a floating or latent initial /k/.[11]

The irrealis infix often combines with complex stems, including—as relevant to this section—compounds. A compositional compound, like that in (18a), remains intact when the infix intrudes between the members of the compound, (18b) (H:81).

(18) a. bilaw-chi:-ya
fish-excrements-POSS
'fish excrements'
b. bilaw<*a'*>-chi:-ya
<IRR>fish-excrements-POSS
'there are no fish excrements'

If the infix were to semantically (and not just linearly) intrude in the compound, we might expect (18b) to have a meaning something like 'there are excrements of no fish', where the negative existential meaning scopes over just one member of the compound. Instead, we find the irrealis clearly scoping over the whole compound in the meaning provided in H.[12]

The interpretation of a compound that is idiomatic also survives the intrusion of the irrealis infix. The example in (19a) shows a compound that consists of the verb 'work at' and the noun 'utensils', producing the regular way to express the verb meaning 'work' in Movima. This idiomatic interpretation, 'work,' persists when the compound combines with the irrealis morpheme, (19b).

(19) a. tij-ka:rim
work.at-utensils
'work' (verb) (H:351)
b. kas tij<*a'*>-ka:rim
NEG <IRR>work.at-utensils
'nobody is working/at work' (H:439)

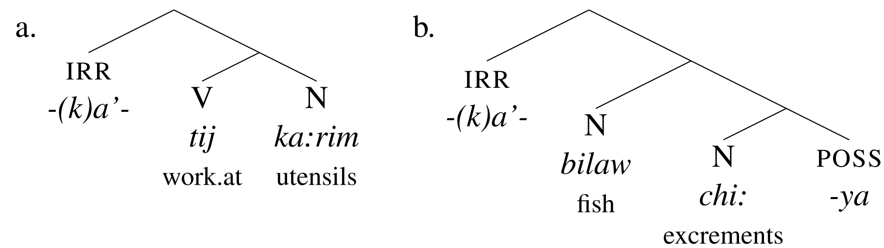
[11] Analyzing the variation between *-ka'-* and *-a'-* as a suppletive alternation (i.e., treating them as different exponents) is hypothetically possible, but would run counter to the finding in Kalin 2022a that infixes do not undergo suppletive alternations in their surface (infixed) positions. Note that there are other affixes that alternate between a vowel-initial form and an otherwise-identical consonant-initial form, with that consonant being /k/ for some other affixes as well (H:39). But, glottal stop is the more general hiatus-avoidance strategy in Movima (H:43–44). The appearance of the *k* preempts glottal insertion.

[12] Although no other meaning of (18b) is provided in H, it is of course possible that there is an alternative scopal meaning. If so, given the ability of irrealis to scope over the whole compound, I would assume that a distinct structure underlies any alternative meaning. The same holds for the compound in (19b), if an alternative meaning were to be possible.

As observed by H:81: the irrealis infix's "semantic scope is over the entire word, not only over the preceding segment".

If the irrealis infix were to semantically (and not just linearly) intrude in the compounds in (18b) and (19b), we would expect their meanings to be disrupted and any idiomatic meaning to be lost, just as in (14b) and (15b). The survival of compound meaning supports the following structures for the words in (18b) and (19b), assuming some kind of mapping between meaning and structure:

(20)



In both structures in (20), the compound is a structural constituent. The infix, even though it (ultimately) linearly intervenes in the compound, is morphosyntactically outside of the compound, based on both its semantic scope and its status as a high inflectional element. I place the infix in a left branch, because its post-first-foot position is oriented with respect to the beginning of its stem, but nothing specific hinges on the mode of basic morphosyntactic linearization.[13] What is important is that the linear interruption of the infix in the compound does not disrupt the interpretation of the compound, as expected given the constituency of the compound (see also §2.1).

3.1.2 Overall findings: Semantic relationships

The survival of semantic interpretation (idiomatic and compositional) in the stem of an infix is consistent throughout the entire sample of infixes in this study, and will be evident in many of the examples in the remainder of the paper. Importantly, this finding holds true not just for intruding inflectional morphemes like the irrealis in Movima, but also for intruding derivational morphemes, as will be seen for example in Nancowry and Hunzib below. It also holds true for all of the excluded case studies mentioned in §1.2.1. Infixes, even when appearing intermorphemically, are not semantically interpreted in their phonological location; rather, they are interpreted as *external* to their stem. As discussed in §2.1, this particular result is not surprising (after all, there's a reason we call them infixes!), but it is still an important part of the full picture that I will build.

3.2 Morphosyntactic relationships (in the stem of an infix) survive infixation

I use the term "morphosyntactic" to scope over phenomena that look like they involve an interaction of morphology and syntax (structure), for example, relationships of selection and intermorpheme compatibility. Such relationships and interactions are highly local and sensitive to interruption.

[13] Throughout the paper, I draw structures such that they can be read left-to-right to get the right morpheme order, with the exception of the infix which may displace from its morphosyntactic position.

Consider category selection, which requires morphosyntactic adjacency. The word *falsifiable*, (21), contains two suffixes, *-ify* (which selects for an adjective and derives a verb) and *-able* (which selects for a verb and derives an adjective).

(21) [[[false $]_{\text{Adj}}$ -ify $]_{\text{V}}$ -able $]_{\text{Adj}}$

Both suffixes satisfy their selectional needs under adjacency—for *-ify*, this is under adjacency with its adjectival stem *false*, and for *-able*, this is under adjacency with its verbal stem *falsify*. The important (though also banal) observation here is that the suffix *-able* cannot appear between *false* and *-ify* (**false-able-ify*) as this would both disrupt the selectional relationship between *-ify* and *false* and fail to satisfy *-able*'s selectional restrictions. Morphosyntactic selection requires locality.

With respect to compatibility, consider the comparative suffix *-er* in English, which is picky, combining only with certain adjectives;[14] otherwise, the comparative is expressed periphrastically with *more*. One adjective compatible with *-er* is *simple*, (22a). If, however, *simple* is embedded in a morphologically complex adjective, like *simplified*, then the *-er* suffix is no longer compatible, (22b).

(22) a. simpler
b. *simplifieder (cf. ✓more simplified)

Another adjective compatible with *-er* is *green*, (23a), and again, acceptability of *-er* is lost across an intervening adjectival suffix, *-ish*, (23b).

(23) a. greener
b. *greenisher (cf. ✓more greenish)

The point being made here is that when other morphemes intervene between *-er* and a root that is compatible with *-er*, this compatibility is voided, and the periphrastic strategy is required.

The question I'll now pose is as follows: what happens when the relevant local morphosyntactic relationship is in the stem of an infix, and the infix appears between the morphemes that are a part of that relationship? The answer is that—just as with semantic interpretation (§3.1)—morphosyntactic relationships are unaffected by an intruding infix. This is again as expected by all theories (§2.1). This section returns to Movima, §3.2.1, then introduces a new case study of Eton, §3.2.2, before zooming out to the overall picture, §3.2.3.

[14] Note that while it is true that *-er* generally combines with mono- or disyllabic adjectives, this is neither a necessary nor sufficient condition for predicting its compatibility with a given adjective, cf. the ungrammatical **iller* or **pinker*, but the grammatical *unhappier*. See discussion in Bobaljik 2012:Ch. 5.5. I therefore assume there is some kind of lexical specificity involved in the compatibility of *-er* with an adjective.

Charles Yang points out (p.c.) that *-er* sometimes seems to "select for" complex stems like *lucky*. A quick perusal of adjectives derived with *-y* suggests (by my intuition) that all are compatible with *-er*, even infrequent ones, for example, *soapy/soapier, milky/milkier*, and novel ones, for example, *wuggy/wuggier*. I would therefore suggest that the compatibility of these derived adjectives with *-er* stems from *-er*'s compatibility with suffix *-y*, rather than with the complex form as a whole.

A final note here—pointed out by an anonymous reviewer—is that with the right kind of context, acceptability of *-er* can be coerced even with adjectives it does not normally go with. I leave open the question of what governs this phenomenon.

3.2.1 Movima: Morphosyntactic relationships

Recall from §3.1.1 that in Movima the irrealis morpheme is realized infixally, as *-(k)a'-*, which appears after the first foot of its stem. I'll focus on two examples of the irrealis intruding, linearly, between morphemes that stand in a special morphosyntactic relationship.

The first special morphosyntactic relationship involves the multiple event (MLT) suffix *-ka*, underlined in the examples, which *only* combines directly with roots (H:78), for example, as it does in (24a). (In this example, MID is a middle voice morpheme that is realized as infixing reduplication before the final syllable of the stem (H:§8.3.3), and *ba* 'round' is a classifier indicating the shape of an internal argument (H:§9.2.6).) The example in (24b) shows that the irrealis can surface between MLT *-ka* and the root, without inducing ungrammaticality.

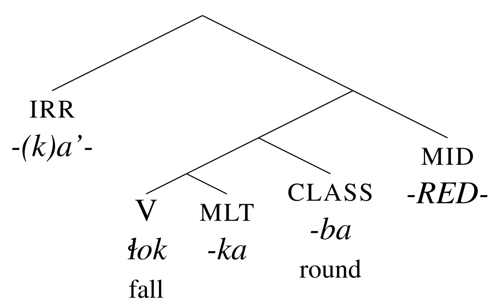
(24) a. sul-ka-<ba:>ba
entangle-MLT-round<MID>
'be all entangled' (H:153)

b. ɬok<*a'*>-ka-<ba:>ba
fall<IRR>-MLT-round<MID>
'may fall down (by themselves)' (H:79)

In fact, the irrealis infix is the *only* morpheme that can appear in an intervening position between MLT and a root; in all other cases, MLT *must* be immediately surface-local to a root. As noted by H:79, "this shows that the irrealis marker is inserted after the other affixes have been attached to the base."

Given that the MLT suffix must be morphosyntactically immediately local to a root, and that the irrealis morpheme is the outermost affix (see §3.1.1), this supports a structure like that in (25) for (24b):

(25)



Just as in (20), the irrealis is structurally higher than all other affixes, and so the necessary local morphosyntactic relationship between MLT and the root still holds. Recall, also as in (20), that I place the infix morphosyntactically in a left branch to reflect that its pivot is at the left edge of its stem.

Another morphosyntactic process that survives infixation in Movima is object incorporation. Movima has a productive process where a transitive verb "incorporates a nominal element that would represent [the absolutive argument] if occurring outside the verb" (H:§7.7). This process is illustrated in (26) (H:368), where the underlined internal argument in (26a) is incorporated into the verb in (26b).

(26) a. wul-na =n kis sani:ya
sow-DIR =2ERG ART.PL melon
'You sow melon.'
b. ij wul-a-sani:ya
2ABS sow-DIR-melon
'You sow melon.'

Object incorporation is undisturbed when the irrealis infix intervenes between the incorporating object (underlined below) and the verb (H:79):

(27) a. iń jiɨ-a:-pa[15]
1ABS grate-DIR-manioc
'I grate manioc.'
b. iń jiɨ-a<*ka'*>-pa
1ABS <IRR>grate-DIR-manioc
'I'll grate manioc.' (H:79)

The example in (27a) shows a simple incorporated object, while (27b) shows the persistence of this incorporation across the infix.

Morphosyntactic relationships of compatibility and incorporation survive across an intruding infix in Movima. I will have more to say about the nature of object incorporation in Movima in §4.2, where I argue it provides support for a post-syntactic model of infixation.

3.2.2 Eton: Morphosyntactic relationships

Eton is a Bantu language spoken in Cameroon (Van de Velde 2008; henceforth V). Eton is synthetic and features an abundance of tonal morphology. I introduce the infix of interest and then turn to the relevant morphosyntactic relationship at hand.

In Eton, there is a morpheme called the "G" morpheme (its name being a reference to the presence of a *g* in all its phonological exponents) that derives the verb stem in a number of tense/aspect combinations (V:Ch. 7.2.2).[16] There are several allomorphs of the G morpheme, and we'll be interested in the one that combines with stems of shape CVCV. The allomorph of interest is an infix of the form *-Lg-*, where L is a floating low tone that docks to the left, (28).[17]

(28) a. bèbè (look.at) + *-Lg-* (G) → bèb<*g*>è (V:246)
b. kódò (leave) + *-Lg-* (G) → kôd<*g*>ò

[15] The form of the root meaning 'manioc' here is *pa*, which is a "bound lexical element" (H:72–73), with the same meaning as the free form of this root, *chinaɫa*. I comment further on this in fn. 50.

[16] "The G-form of the verb is used in the Hesternal past perfective, in the Past imperfective, the Relative imperfective, and in the participial involved in the formation of past imperfective verb forms" (V:245).

[17] The tonal changes that happen upon the addition of the G morpheme are predictable from general tonal rules in the language (V:57–58). In (28a), the floating L of the infix is deleted when it follows an identical tone. In (28b), the low tone docks to the left to create a falling tone on the first syllable; if *kódò* were uninflected, it would surface as *kódô*, with high-tone spreading to the final syllable (V:59).

As can be seen in (28), the infix appears before the final vowel of its stem, and can go inside of roots.

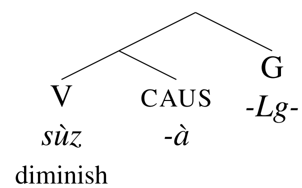
While stems of the shape CVCV (i.e., stems that take the *-Lg-* infix) may be monomorphemic, like those in (28), they may also be bimorphemic. In particular, they may consist of a CVC root with the causative suffix *-à*. The causative suffix, in turn, is very picky—the synthetic (suffixal) causative strategy is limited to a restricted and arbitrary class of roots (V:Ch. 4.3.1–2). For all other verbs, the causative must be expressed periphrastically. Notably, when the G morpheme combines with a stem that includes this picky causative suffix, the infix appears between the root and the suffix; nevertheless, the infix does not disrupt root+causative compatibility, (29).

(29) sùz-à (diminish-CAUS) + *-Lg-* (G) → sùz<*g*>-à (V:246)

If the infix were to disrupt compatibility, we'd expect a retreat to the periphrastic strategy, counter to fact. This, in turn, shows that the compatibility relation of the causative with a particular root is established *prior* to the intrusion of the infix, or at a level of representation where the infix does not intervene.

Given the facts about the G morpheme, I posit the following structure for (29):

(30)



Counter to the linear order in (29), where the G morpheme is between the root and the causative morpheme, the causative morpheme is in fact structurally closer to the root than the G morpheme. This morphosyntactic locality between the root and the causative morpheme is why the root+causative compatibility relationship survives.

3.2.3 Overall findings: Morphosyntactic relationships

What I showed above for Movima and Eton could be illustrated with every case study in the sample: morphosyntactic relationships in the stem of an infix always survive the intrusion of the infix. Further examples that will be seen in sections below include a nominalizer intervening between a causative morpheme and its stem (Nancowry, §3.3.2, as well as Katu, §3.3.3) and a verbal plural marker intervening between a verbalizer and its nonverbal stem (Hunzib, §3.4.2, §3.5.2). These examples illustrate clearly that morphosyntactic selection is maintained despite the intrusion of an infix.

The excluded case studies (§1.2.1) are also all in line with the findings in this section. One example that bears mentioning—in terms of morphosyntactic idiosyncracy surviving across an infix—emerges with expletive infixation in English. In Ahn and Kalin 2018, we show that the exceptional accusative case that is found in some Mainstream English reflexives (e.g., accusative *him* in *him-self*, cf. genitive *my* in *my-self*) can survive in the presence of the expletive (e.g., *him-<fucking>self*). The survival of the accusative in such cases is crucially unlike other (noninfixal) disruptions of the reflexive—with other kinds of disruptions, the exceptional accusative is lost (e.g., *his sweet self*, **him sweet self*). The analysis we offer in Ahn and Kalin 2018 capitalizes on the timing of infixation of the expletive, taking this disruption to be able to occur after the case idiosyncrasy is conditioned.

The implication across the board is that infixes do *not* intervene at a structural level, even when they have an intermorphemic position. As discussed in §2.1, and like the finding in §3.1, this finding is solidly in the realm of "not surprising"—indeed, it's a relief that canonical infixes *do* behave this way, because the operation of infixation is often invoked when trying to explain morphosyntactic nondisruption or counter-scopal orders (e.g., Rice 1987, Orgun 1996, Crysmann 2000).

3.3 Suppletive allomorphy (in the stem of an infix) survives infixation

Suppletive allomorphy describes a situation where the operation of exponence must make reference to the environment of a morpheme in order to choose the right exponent. (See, e.g., discussions in Mel'čuk 1994, Veselinova 2006, Bobaljik 2012, Kalin 2022a.) Just like semantic interpretation (§3.1) and morphosyntactic compatibility (§3.2), usually suppletive allomorphy is disrupted under intervention.

Consider lexically conditioned suppletive allomorphy of the English plural morpheme: plural is usually exponed as *-s* (phonologically /-z/), but has a number of suppletive alternatives that depend on the root/stem, like *-ren* (in *children*), *-en* (in *oxen*), -∅ (in *fish*), and *-i* (in *alumni*). If we take a noun root that conditions a special form of the plural, (31a)/(32a), and derive a more complex noun from it (adding e.g. *-y* or *-let*), plural marking then retreats to the default *-s*, (31b)/(32b):

(31) a. one fish, two fish-∅
b. one fish-y, two fish-ie-s (*two fish-y-∅)

(32) a. one ox, two ox-en
b. one ox-let, two ox-let-s (*two ox-let-en)

The suppletive allomorphs of the plural seen in (31a) and (32a) (-∅ and *-en*) do not survive across intervening affixes.

The same is true of grammatically conditioned suppletive allomorphy. In English, *good* and *bett* alternate as exponents of the same morpheme, the root meaning 'good'. The exponent *bett* appears in the context of comparative morphology (*-er*), and *good* appears elsewhere, (33a). The *good/bett* alternation is blocked by any intervening morphology (*-ly* here), with *good* surfacing even in the comparative form, (33b).

(33) a. good, bett-er
b. good-ly, good-li-er (*bett-li-er)

Grammatically conditioned allomorphy is usually disrupted by an intervening affix.

And finally, phonologically conditioned suppletive allomorphy is also disrupted under intervention. We can observe this in the alternation of the indefinite article between *a*, /eɪ/, and *an*,

/æn/, shown in (34a). (For discussions of *a/an* allomorphy and why it should be treated as suppletive, see Smith 2015, Pak 2016.) If anything intervenes between the indefinite article and the noun root, it is always the *closest* phonological material that conditions the choice between *a* and *an*, (34b), never some further away phonological material (e.g., that of the root).

(34) a. a banana, an apple
b. an un-banana, a non-apple[18]

Suppletive allomorphy—no matter what conditions exponent choice—is conditioned locally, and is generally interrupted under intervention.[19]

Does the intervention of an infix have the same disruptive effect on suppletive allomorphy? As discussed in §2.2, predictions differ across theories here, with once-and-for-all approaches to exponence predicting disruption, and bottom-up approaches predicting nondisruption; multistage approaches predict nondisruption only if there is a stage boundary between the infix and its stem. The finding, consistent across the sample, is that infixes are invisible for suppletive allomorphy within their stem. In other words, infixes are *not* interveners in the locality required by suppletive allomorphy, no matter the type of conditioning environment. This finding supports bottom-up exponence across the board, for all types of exponence.

In §3.3.1 we'll return to the Palauan example from the introduction of this paper, drawing on a case study by Embick (2010:104–107). In §3.3.2, I turn to Nancowry, utilizing one of my own earlier case studies (Kalin 2023). The overall findings are discussed in §3.3.3.

3.3.1 Palauan: Suppletive allomorphy

Embick (2010:104–107) offers a case study of Palauan infixation and allomorphy. The discussion here largely follows Embick's, though the examples are drawn from other works. What Embick shows is that intervention of the past tense infix does not block allomorphy of the verb marker. I'll introduce both morphemes in turn below.

As mentioned briefly in the introduction, the past tense marker in Palauan is an infix, *-il-*, that appears after the first segment of its stem. Its infixal nature is clear from observing bare verb roots that combine with the past tense, like those in (35) (Josephs 1975:190):[20]

(35) a. *-il-* (PST) + siik (look.for) → s<*il*>iik
b. *-il-* (PST) + ker (ask) → k<*il*>er

The past tense marker therefore fits the profile of a canonical infix.

Verbs in Palauan usually bear a "verb marker" (VM). As put by Josephs (1975:148): "It is very difficult to define or specify the meaning of the verb marker; rather, the best we can do is to say

[18]These are obviously odd words, but it's not too hard to cook up a context where they'd be interpretable.

[19]For discussions of exactly what the relevant locality condition on suppletive allomorphy is, which may vary by type of allomorphy, see for example Embick 2010, Bobaljik 2012, Merchant 2015, Moskal 2015, Choi and Harley 2019, Paparounas 2024.

[20]Palauan verb morphology is incredibly complex, and I cannot do justice to this complexity here. I refer the interested reader to Josephs 1975:Ch. 5–7 and Nuger 2010 for extensive discussions.

that the verb marker simply functions to mark or identify a particular word as a verb." Nuger 2010 analyzes this affix as instantiating *v* (which has a number of other flavors and realizations). The verb marker has two suppletive allomorphs, *m(ə)-* and *o-*. The overwhelming majority of verbs take *m(ə)-*, (36a). All bilabial-initial verb stems take *o-*, (36b), as do a small list of other verbs, like that in (36c) (Josephs 1975:Ch. 6).

(36) a. VM + dasaʔ (carve) → mə-dasaʔ (Flora 1974:99)
b. VM + baloʔ (shoot) → o-baloʔ (Flora 1974:100)
c. VM + siik (look.for) → o-siik (Josephs 1975:133)

While the appearance of *o-* in (36b) is phonologically conditioned (and may even be due to (morpho)phonology rather than suppletion), the appearance of *o-* in (36c) is surely suppletive, and the choice of *o-* in such cases must be lexically conditioned.

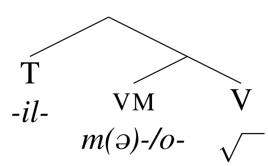
Of course the question now is what happens when a VM-marked verb takes the past tense infix. The VM allomorph that appears in the nonpast form (without the intrusion of the infix) survives in the past tense, across the infix, (37) (cf. these same verb roots in (36)):

(37) a. m-<*il*>dasaʔ (PST + VM-carve) (Flora 1974:100)
b. o-<*il*>baloʔ (PST + VM-shoot)[21] (Flora 1974:101)
c. o-<*il*>siik (PST + VM-look.for) (Josephs 1975:133)

The relevant generalization is as follows. If the elsewhere allomorph of the verb marker, *m(ə)-*, would have appeared in the noninfixed form (like it does in (36a)), then this allomorph continues to appear in the presence of the past tense infix (as in (37a)); if *o-* would have appeared due to phonological conditioning or lexical conditioning in the noninfixed form (like it does in (36b-c)), then this allomorph persists as well in the past tense (as in (37b-c)). The overall observation here is that allomorph choice—whether phonologically conditioned or lexically conditioned—survives the intrusion of the infix.[22]

Given the low, categorizing(-esque) function of the verb marker, and the high inflectional nature of tense, I assume the following structure for verbs bearing both affixes:

(38)



Since the verb marker is structurally closer to the root, the observations above can be understood if the exponent of the verb marker is chosen before exponence/infixation of *-il-*. This would be true in bottom-up approaches to exponence; it would also be true in multistage approaches where the verb marker is exponed at an earlier stage than tense, which is reasonable to posit in this case given the distinct morphosyntactic domains of the affixes, one high and one low.

[21] Sequences of *o* followed by *i* merge to *u*, producing *ulbaloʔ, ulsiik* (Josephs 1975:590).

[22] If it turns out that cases like (36b) are better treated (morpho)phonologically, then this would still be compatible with my overall findings; see §3.4.

3.3.2 Nancowry: Suppletive allomorphy

The Austroasiatic language Nancowry (Radhakrishnan 1981, henceforth R) furnishes two further cases of suppletive allomorphy surviving the intrusion of an infix (Kalin 2023). The case I'll illustrate first involves the causative morpheme—which has suppletive allomorphs—and the instrumental nominalizer infix *-in-* as the intruder. The second case involves double causatives, with the outer causative as the intruder.

The first relevant morpheme is the causative, which has two prosodically conditioned suppletive forms (one of which is itself an infix), *ha-* and *-um-*. These allomorphs distribute based on the size of the stem: *ha-* appears with monosyllabic stems, (39a), and *-um-* with disyllabic stems, (39b). The causative is underlined in the following examples.

(39) a. CAUS + kuãt (curve) → ha-kuãt 'to hang, to hook' (R:96)
b. CAUS + paloʔ (loose) → p<um>loʔ 'to loosen' (R:150)

Notice in (39b) that *-um-* seems to swallow up the first vowel of the stem it combines with. In Kalin 2023, I propose that this is because *-um-*'s pivot is actually the first vowel; when the infix takes a post-first-vowel position, it creates vowel hiatus, which is disallowed in unstressed syllables in Nancowry. Vowel hiatus is resolved through deletion of the first stem vowel.

The instrumental nominalizer infix *-in-*, italicized in the examples, is a derivational affix that combines with verbs and derives instrument nouns:[23]

(40) a. *-in-* (INOM) + calɯak (swallow) → c<*in*>lɯak 'a throat' (R:146)
b. *-in-* (INOM) + tikoʔ (prod) → t<*in*>koʔ 'a prod' (R:97)

Just like with *-um-* in (39b), the first vowel of the stem disappears under infixation; this can be accounted for in the same way: *-in-* is placed after the first vowel, and that vowel is subsequently deleted to resolve hiatus.

Verbs bearing a causative prefix can be nominalized by the instrumental nominalizer. When this happens, the choice of suppletive allomorph for the causative, (39), is unaffected. So, for example, *ha-* survives in (41):[24]

(41) h-<*in*>kuãt (INOM + CAUS + curve; 'a hook') (R:96)

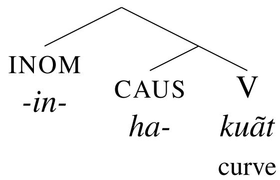
Recall that *ha-* only appears with monosyllabic stems, (39). In (41), looking naively just at the surface form, it *looks* as though the stem that follows *h(a)-* is disyllabic (*inkuãt*). But, nevertheless, the causative allomorph *ha-*—the one that would have been chosen in the absence of the infix, with a monosyllabic stem—persists.

The meaning of (41), as well as the selectional requirements of the morphemes, supports the structure in (42).

[23] The instrumental nominalizer exhibits suppletive allomorphy, but this will not be of interest here.

[24] I do not give an example of the *-um-* allomorph, (39b), surviving infixation because infixation of *-um-* followed by infixation of *-in-* actually results in the surface-disappearance of *-um-*; in Kalin 2023, I show that this can be explained by predictable phonological/phonotactic processes within the language.

(42)



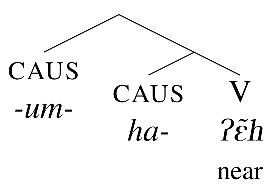
Just like in understanding the Palauan case study above, here the survival of suppletive allomorphy naturally follows from a bottom-up theory of exponence, where the causative morpheme is exponed before exponence/infixation of the higher morpheme. Multistage exponence is also a plausible analysis here, as there could be a stage boundary—where exponence is computed—demarcating the verbal domain from the nominal domain.

A second case from Nancowry comes from double causatives, like that in (43).

(43) h-<*um*>ʔɛ̃h (CAUS + CAUS + near; 'to cause to approach') (R:85)

Double causatives involve the stacking of two causative morphemes, both of which are semantically interpreted. The outer causative morpheme (italicized) is the infixal intruder, into a stem that contains another causative (underlined). Exponent choice for the lower causative survives across the intruder, just like when it was the nominalizer intervening, (41) above. The structure of the double causative in (43) is given in (44):

(44)



While the result is the same, this finding is more automatically accounted for in a bottom-up approach to exponent choice, rather than a multistage approach, since both causatives are presumably in the same verbal domain.

Taken together, Nacowry's two examples of suppletive allomorphy surviving across an infix are most easily accounted for in a bottom-up approach to exponence.

3.3.3 Overall findings: Suppletive allomorphy

In my sample, there is one other case of an infix intruding in an allomorphic relationship in its stem. In Katu (Austroasiatic; Costello 1991, 1998), lexically conditioned allomorphy of the causative morpheme (as *pa-* most commonly, but also *pi-, ka-* or *ta-*, depending on the root) survives the intrusion of a nominalizing infix, *-r-*. For example, the causativized verb *ka-chêt* (CAUS-die, 'kill') is nominalized as *ka-<r>chêt* ('dead'), with the choice of the lexically conditioned allomorph *ka-* persisting. Thus in all cases in the sample, suppletive allomorphy survives infixation.

Since my sample does not furnish many examples of suppletive allomorphy in the relevant environments for testing (non)survival across an infix, I will briefly comment on the other infix-adjacent case studies that I excluded in this paper (§1.2.1). Two are relevant: expletive infixation and the Turoyo past tense morpheme.

Turoyo's infix-like past tense morpheme (see §1.2.1) can intervene in two relevant places: (i) between a dative marker that undergoes grammatically conditioned suppletive allomorphy and the trigger of this allomorphy (agreement morphology), and (ii) between an agreement morpheme that undergoes phonologically conditioned suppletive allomorphy and the trigger of this allomorphy (again found in adjacent agreement morphology). In both cases, the past tense marker does *not* disrupt the allomorphy, in line with the findings above. For details, see Kalin 2020. Relevant here is that these interactions are all in a high inflectional domain, between tense and agreement, and so a bottom-up approach explains these findings more easily than a multistage one, which would require at least one stage boundary within this set of morphemes, possibly two, to capture the findings.

With respect to the excluded case of so-called expletive infixation (see §1.2.1), there is one potentially relevant environment where *fucking* can surface, namely, when it intervenes between a prefix expressing negation (*in-, un-, non-, a-*) and its adjectival stem. Assuming that these negative morphs can be treated as lexically conditioned suppletive allomorphs of a single abstract negative morpheme,[25] then this provides another example of such allomorphy surviving across an infix-like element:

(45) a. un-comfortable, un-<*fucking*>comfortable
b. non-sensical, non-<*fucking*>sensical
c. a-moral, a-<*fucking*>moral
d. in-variant, in-<*fucking*>variant

For all the examples in (45), the idiosyncratic pairing of negative morph and adjectival stem survives across the expletive. This again follows automatically from bottom-up exponence, and potentially also from multistage exponence, assuming a stage boundary between the expletive and its stem.

In all relevant (and potentially relevant but excluded) cases that I have been able to find, infixes consistently fail to interrupt relationships of exponence in their stems; in other words, suppletive allomorphy (in the stem of an infix) always survives infixation. This is the first truly informative result of this study (see §2.2)—under once-and-for-all theories of exponence, the intervention of an infix should *block* allomorphic relationships, especially phonologically conditioned ones. Since, to the contrary, these allomorphic relationships uniformly *survive* infixation, this shows that infixation must happen *after* the infix's stem has been fully exponed, that is, after the morphemes in the stem of the infix have been given phonological forms. (Or, equivalently, what this shows is that exponents must be present at a representational level where the stem of an infix excludes the infix.) This result follows automatically from a bottom-up theory of exponence. While this result is also potentially compatible with a multistage theory of exponence, this type of approach would require there to always be a stage boundary between the infix and its stem in all of my case studies, which is

[25] This analysis is not totally straightforward to defend, as these morphs are not in complementary distribution for all stems, and (with some lexical items) they seem to convey slightly different meanings. For extensive discussion and relevant citations, see Bauer et al. 2013. If it turns out that suppletive allomorphy is not the right way to think about these alternations, then this case would be better categorized as part of the domain of (morpho)syntactic compatibility, §3.3.

more plausible in some cases than others (as noted above). And, a multistage theory of exponence also predicts that there should be cases of *disruption* of suppletive allomorphy by an infix (see (13)), an empirical profile which is fully absent from my (albeit small) sample.

I conclude from the findings in this section that they support bottom-up computation of exponence. This conclusion is strengthened by the findings in the next section, which show that even some phonology must have happened in the stem of the infix prior to infixation.

3.4 Morphophonological interactions (in the stem of an infix) survive infixation

I use the terms "morphologically conditioned phonology" and "morphophonology" interchangeably to pick out a class of phenomena that sits in between suppletive allomorphy (§3.3) and fully regular, word-level phonology (§3.5). Elaborating on the discussion at the beginning of §2.2.2, morphophonology is when there is a phonological alternation that can be described straightforwardly as a phonological change (rather than a suppletive one), but that is morphologically idiosyncratic in some way. As Inkelas (2014:Ch. 2) puts it, "morphologically conditioned phonology is the phenomenon in which a particular phonological pattern is imposed on a proper subset of morphological constructions [...] and thus is not fully general in the word-internal phonological patterning of the language." An example given by Inkelas (p. 10) comes from Mam Maya (Inkelas cites England 1983, Willard 2004): roots with long vowels undergo vowel shortening in the context of a few particular suffixes.

The idea that there can be morphologically conditioned and/or morphologically restricted phonology is a controversial one. Some embrace it, taking a what-you-see-is-what-you-get approach, either positing (i) a "readjustment" rule that enacts a phonological change in a morphologically restricted way (e.g., Embick and Shwayder 2018) or (ii) a specific phonological constraint (or constraint ranking) that applies to a particular morphological element or in a particular morphological environment (e.g., Inkelas 2008, Sande et al. 2020). Others are committed to relegating all apparent cases of morphophonology to either regular phonology (e.g., Newell 2021, crucially with the help of more abstract phonological representations) or suppletive allomorphy (e.g., Merchant 2015). Regardless of its theoretical treatment, though, morphophonology is still a useful descriptive category for the apparent behavior of a set of "in between" empirical phenomena, and it is this descriptive category that I appeal to here. And, we will see that it *does* behave differently than word-level phonology does (§3.5).[26]

Morphophonology, as defined above, generally obeys strict locality.[27] As a simple example, consider the fricative voicing alternation found in plural environments for some nouns in English:

(46) (47) a. lea[f], lea[v]es
b. hou[s]e, hou[z]es
c. mou[θ], mou[ð]s

[26]This class of phenomena also behaves differently from suppletive allomorphy, as I show extensively in Kalin 2022a.

[27]See Embick and Shwayder 2018 for a discussion of locality and how it might be somewhat different for different types of morphophonological processes.

This alternation is not general to all noun roots (cf. *ree[f]*, *ree[f]-s*, **ree[v]-es*),[28] and is also conditioned only in a plural environment (cf. the environment of possessive *-s*: *the lea[f]'s stem*, **the lea[v]'s stem*). The phonological process of voicing is therefore doubly morphologically restricted. Turning to intervention: in the usual case, when another morpheme intervenes between plural marking and a relevant noun root, voicing of the root-final consonant no longer occurs:

(48) a. lea[f]let, lea[f]lets (*lea[v]lets, cf. (47a))
b. mou[θ]ful, mou[θ]fuls (*mou[ð]fuls, cf. (47c))

The special roots that undergo voicing of their final fricative only do so when the plural suffix is immediately local, as in (47). Put another way, morphophonology is generally bled by intervention.

So far, we have seen that semantic interpretation, morphosyntactic compatibility/selection, and suppletive allomorphy all survive the intrusion of an infix (§3.1–3.3). What about morphophonology? As discussed in §2.2, predictions differ across theories: once-and-for-all approaches to morphophonology predict disruption, and bottom-up approaches predict nondisruption, with multistage approaches in between. The finding here is that infixes are invisible for morphophonology within their stem. This finding supports the computation of morphologically conditioned phonology from the bottom up.

I begin by returning to Movima, §3.4.1, and then turn to a case study of Hunzib undertaken in Kalin 2022b, §3.4.2. Overall findings are presented in §3.4.3.

3.4.1 Movima: Morphophonological interactions

Recall from §3.1.1 that Movima has an irrealis morpheme that is realized as an infix, *-(k)a'-*, which follows the first foot of its stem (Haude 2006). Movima is also a language rich in morphophonological processes, one of which will be of interest in this subsection: reduplication.

While reduplication sometimes clearly realizes a morpheme in Movima, reduplication also appears to be morphophonological in some cases, that is, triggered as a phonological process rather than due to a segmentally empty exponent (H:§3.7). For example, in nominal compounds built from a verb root, the verb root—which is always the left member of the compound, the non-head—undergoes prefixing reduplication of its first two moras (H:200–202), (49) (reduplicants underlined).

(49) a. dan (chew) + so (chicha) → <u>dan-</u>dan-so 'chicha made of chewed maize'
b. jo: (warm.up) + mi (water) → <u>jo:-</u>jo:-mi 'warm water'
c. sam (twist) + di (long.thin) → <u>sam-</u>sam-di 'rope'

Since it is not clear what the reduplicant would be *realizing* in such compounds, I take reduplication here to be plausibly morphophonological, following Haude. (And, even if it turns out that reduplication in (49) is due to the presence of a segmentally empty exponent, the mechanism responsible for copying the stem material is presumably still morphophonological in nature.)

[28] There is also cross-dialectal variation in terms of which noun roots undergo this alternation; see for example MacKenzie 2018.

Another context for morphophonological reduplication is when a relational noun root is used nonrelationally, without an inalienable possessor (H:208–212). In this case, the absolute state suffix *-kwa* is required, for example, (50).

(50) tolej (branch) + -kwa (ABSL) → tolej-kwa 'branch'

For roots larger than one syllable, like that in (50), this is all that is involved in deriving a nonrelational nominal. However, for some (lexically arbitrary) monosyllabic roots, they must undergo prefixing CV-reduplication in this context, in addition to taking the *-kwa* suffix (p. 209), (51) (reduplicants underlined).[29]

(51) a. ba (fruit[30]) + -kwa (ABSL) → ba-ba:[31]-kwa 'fruit'
b. di (grain) + -kwa (ABSL) → di-di-n[32]-kwa 'seed'

Because *-kwa* can occur with or without accompanying reduplication, and the set of roots undergoing reduplication is (to some degree) lexically arbitrary, I take reduplication to be morphophonological in these cases as well.

For the purposes of the current investigation, the question is what happens when morphophonological reduplication—e.g., of the type found in certain compounds, (49), or in deriving certain nonrelational nouns, (51)—is triggered in the stem of the irrealis infix. And the answer is that this reduplication remains intact despite the intrusion of the infix: reduplication is still required, and the form of the reduplicant remains unchanged. Sometimes the post-first-foot location of the infix places the infix between the reduplicant and its base, as in (52a), and other times after the reduplicant+base, (52b) (H:80).

(52) a. *-(k)a'-* (IRR) + sam-sam-di ('rope', (49c))
→ sam-<*a'*>sam-di 'there is no rope'

b. *-(k)a'-* (IRR) + ba-ba:-kwa ('fruit', (51a))
→ ba-ba<*ka'*>-kwa 'there is no fruit'

Importantly, in both cases, it's clear that morphophonological reduplication in the stem is oblivious of the infix: first of all, reduplication must have already happened for the infix to find its correct (attested) location; and second, infixation does not disrupt the reduplication or affect the shape of the reduplicant in any way. In addition, when considering reduplication of monosyllabic roots in the nonrelational construction, (51), the presence of the irrealis infix does not obviate this (at least

[29]In some cases, this reduplication seems to occur to satisfy prosodic minimality constraints—nouns must minimally have three moras, with moras added by penultimate lengthening (fn. 31) counting towards this minimum—but reduplication does not *only* happen when this is prosodically necessary (H:45). Further, different monosyllabic roots trigger different prosodic augmentation/repair strategies in different contexts (H:205), and so reduplication must involve some morphological idiosyncrasy in any case.

[30]Haude glosses this as 'round' because in other contexts it serves as a shape classifier. But, as she discusses (p. 212), it may be that the shape classifier is actually a semantically bleached version of a noun meaning 'fruit', and so this is how I gloss it here for transparency of meaning.

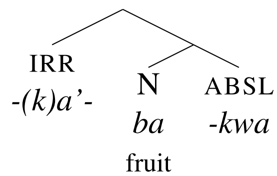
[31]Lengthening here is due to a regular phonological process of penultimate lengthening (H:47–49).

[32]This is a morphophonologically inserted nasal, which will be discussed in §3.5.

partially) prosodically motivated reduplication by adding a syllable, nor by separating the root and the relational suffix.

Given the scope of the irrealis morpheme, I assume a structure like that in (53) for (52b):

(53)



The irrealis morpheme linearly, but *not* morphosyntactically, intervenes between the root and the absolute state suffix. The fact that infixation of the irrealis infix does not obviate morphophonological reduplication in this construction, nor feed into it, shows that reduplication must be triggered before the exponence/infixation of the infix, that is, at the constituent node containing just N and ABSL.

Once again, the data point towards bottom-up computation, this time of morphophonology. The Movima data here are also compatible with multistage computation of morphophonology, if there is (as would be reasonable) a stage boundary between the irrealis morpheme and its stem.

3.4.2 Hunzib: Morphophonological interactions

A second case study featuring morphophonology is found in the Northeast Caucasian language Hunzib (van den Berg 1995; henceforth B), which I discuss in detail in Kalin 2022b. In Hunzib, there is a verbal plural infix, *-á-*, that appears before the final consonant of its stem, (54); this infix has a number of *non*suppletive allomorphs, including *-yá-, -wá-*, and *-á-*.

(54) a. ɑhu (take) + *-á-* (VPL) → ɑ<*á*>hu 'take (PL)' (B:284)
b. ek (fall) + *-á-* (VPL) → e<*yá*>k 'fall (PL)' (B:295)
c. šoše (bandage) + *-á-* (VPL) → šo<*wá*>še 'bandage (PL)' (B:334)

The verbal plural infix combines with verbs, as seen in (54).

There is a productive derivational suffix in Hunzib that forms verbs, *-k'(e)*, a causativizer (underlined below).

(55) a. haldu (white) + -k'(e) (CAUS) → haldu-k' 'make white' (B:301)
b. uƛ' (end) + -k'(e) (CAUS) → uƛ'-k'e 'make end' (B:337)

The alternation between *-k'*, (55a), and *-k'e*, (55b), is at least partially phonological: *e* prevents a complex coda in cases where its stem ends in a consonant, like (55b). However, this is not regular phonology: for resolving complex codas in Hunzib, which vowel is inserted (and where) is idiosyncratic to the affix creating the illicit cluster. For example, the present tense suffix is *-č* after a vowel and *-čo* after a consonant (B:83), and the preterite suffix has the form *-r* after a vowel and *-Vr* after a consonant (B:84), with the underspecified vowel determined by a set of harmony rules that only affect underspecified vowels like this one (B:75). (See further cases and remarks in B:§5.2). I therefore take the alternation in the causative suffix to be best analyzed as a morphologically conditioned phonological process of epenthesis.[33]

When the verbal plural infix combines with a derived verb like one of those in (55), the infix shows up (incidentally) in between the root and the derivational suffix. What happens to morphophonologically conditioned vowel epenthesis, like that in (55b), when the infix intrudes? The answer is that epenthesis proceeds as though the infix weren't there: if there *would not* have been vowel epenthesis, there still is not, (56a) (compare with (55a)); if there *would* have been vowel epenthesis, there still is, (56b) (compare with (55b)).[34]

(56) a. haldu (white) + -k'(e) (CAUS) + -*á*- (VPL) → hald<*á*>-k' (B:301)
b. ɨx (warm) + -k'(e) (CAUS) + -*á*- (VPL) → ɨx<*á*>-k'e (B:308)

Importantly, we might have expected the infix to *bleed* vowel epenthesis, as it creates an environment where the causative morpheme follows a vowel; notice that in (56b), the epenthesized vowel is in some sense superfluous, as there would be no illicit consonant cluster even in its absence. (56b) shows that morphophonological vowel epenthesis must have been determined *prior to* (or in the absence of) the infix.

Once again, a reasonable structure for these words helps to make concrete how to account for the *lack* of intrusion of the infix. Given the derivational nature of the causative affix, which creates the right kind of verbal base for the verbal plural affix, I take (57) to be the structure of (56b):

(57) [[Adj *ɨx* warm] [CAUS *-k'*]] [VPL *-á-*]

If *ɨx* and *-k'* interact morphophonologically in a bottom-up fashion, vowel epenthesis should apply in the constituent that contains just these morphemes, and this epenthesis should not be obviated by later infixation of the verbal plural. While multistage morphophonology could work to account for the findings as well, this would require a stage boundary in between the causative and the verbal plural morpheme, which are both part of low verbal structure.

3.4.3 Overall findings: Morphophonological interactions

The case studies from Movima and Hunzib show that infixes do not disrupt morphophonology in their stems. This finding also holds in two of the excluded case studies (§1.2.1), the Cibemba applicative and English expletives. In Cibemba, morphologically conditioned lenition (triggered by the causative morpheme) famously persists across an intruding ("infixed") applicative morpheme (Hyman 1994, Orgun 1996, Myler 2017, i.a.). And in English, many speakers allow the persistence of *im-* (an assimilated version of the negative prefix *in-*) across an intruding ("infixed") expletive,

[33] This epenthesis could instead be analyzed representationally, for example, with a floating *e* as part of the causative's exponent. The question, then, would be about the timing of when the linking/realization of *e* is determined, which would still crucially need to be bottom-up, as will be shown momentarily.

[34] The final vowel of the root in (56a) is deleted due to a predictable phonological process of hiatus resolution. See discussion in Kalin 2022b:8–9.

as in forms like *im-<fucking>possible*.[35] Assuming that morphophonology is strictly local, all of this means that the stem must have undergone some phonological processes—namely, those that are morphologically idiosyncratic in some way—*before* the intrusion of the infix (or at a representational level that excludes the infix).

The findings in this section support the bottom-up computation of morphophonology (see §2.2), which goes hand in hand with the finding from §3.3 that exponence is also bottom-up. Similar to the discussion in §3.3.3, multistage morphophonology could also work to account for the findings here, but there would always need to be a stage boundary between the infix and its stem; this is a suspicious coincidence across an accumulating set of cases. I therefore take bottom-up morphophonology to be the best-fit type of account for the findings.

The reader might now be wondering whether there is *anything* that fails to survive the intrusion of an infix, and the answer is yes, word-level phonology is interrupted by the intrusion of an infix, as we will see in the coming section.

3.5 Word-level phonology (in the stem of an infix) is lost under infixation

Finally, we turn to phonological processes that are regular in a language, that are not morphologically idiosyncratic, and that apply at the word level. (In §4.1.1 I will return to the question of how regular phonology that is "cyclic", that is, that seems to apply multiple times during the formation of a word, can also fit into the picture.)

Phonological processes are generally conditioned locally. Consider, for example, the well-known case of flapping in many dialects of English, where a *t* or *d* is realized as a flap when it follows a vowel, *r*, or *w* (i.e., following a nonlateral approximant) and precedes an unstressed vowel, as in *odd/odder*, (58a) and *hot/hotter*, (58b).[36] (Flapping is actually a phrase-level process of English, but it behaves like all purely phonological processes in being highly interruptible, and so suffices for the illustration here.)

(58) a. o[d], o[ɾ]-est
b. ho[t], ho[ɾ]-est

If a consonant-initial affix intervenes between *d/t* and the following unstressed vowel in the flapping cases in (58), this bleeds (disrupts) flapping. Consider the addition of the suffix *-ly* in (59), intended to contribute the adjectival meaning "in a(n) X way".

(59) a. o[d]-ly, o[d]-li-est (*o[ɾ]liest)
b. ho[t]-ly, ho[t]-li-est (*ho[ɾ]liest)

[35] Other speakers prefer *in<fucking>possible*. I hypothesize that this contrast is due to whether a speaker has analyzed the *im-/in-* alternation as morphophonological or phonological, in line with the survival of morphophonology (this section) but not phonology (§3.5). There are lots more interesting things to say about *fucking* and morphophonology, for example, about what speakers do with *fucking* in adjectives like *irregular* and *illegal*, which varies across speakers (as *ir-<fucking>regular, i-<fucking>regular*, or simply ineffable). This calls for further investigation.

[36] There is an abundance of literature on English flapping and the exact conditions under which it happens. For many relevant citations and an overview of the major issues, see Elzinga and Eddington 2008.

Although *oddliest* and *hotliest* are marginal words, and while there may be other allophones of *t/d* in this context, the point here is just that flapping would clearly *not* happen when an unstressed vowel, for example, that of *-est* above, does not immediately follow *d/t*. Phonological processes are—in the usual case—bled under intervention.[37]

To see what happens when an infix interrupts a purely phonological interaction, we'll turn to new facets of the case studies from the previous section, Movima and Hunzib, taken up in §3.5.1 and §3.5.2, respectively. Recall from §2.2 that if a phonological process is computed bottom-up (or multistage, at a stage excluding the infix), then it should survive the intrusion of an infix, but if it is computed once-and-for-all (or multistage, at the same stage as the infix), then it should *not* survive the intrusion of an infix. We will see that in Movima and Hunzib, word-level phonology *is* disrupted by the intrusion of an infix, in contrast to all previous types of relationships/interactions considered in this paper, including morphophonology.

3.5.1 Movima: Phonological interactions

In Movima, a robustly evidenced aspect of the word-level phonology is nasal assimilation. H:34 states that "the alveolar nasal phoneme /n/ always assimilates to the following consonant with respect to its place of articulation". Specifically, the alveolar nasal surfaces as: (i) [m] (orthographic *m*) before bilabials, (ii) [ŋ] (orthographic *n*) before velars, and (iii) [n] (also orthographic *n*) before alveolars and vowels.[38] This is illustrated in (60) for the positional root *en* ('stand'). (The relevant nasals are underlined throughout all the Movima examples below. I use orthography except when needing to indicate the presence of a velar nasal.)

(60)
a. en ('stand') + -to (side) → ento 'stand obliquely' (H:34)
b. en ('stand') + -u'-ni (INT-PRC) → enu'ni 'still standing' (H:578)
c. en ('stand') + -poj (CAUS) → empoj 'make stand' (H:34)
d. en ('stand') + -ki (IMP) → e[ŋ]ki 'stand up!' (H:34)

(60c) shows *n* assimilating to a following bilabial, and (60d) to a following velar.

Another context where nasal assimilation is evident is with the so-called "linking nasal" (H:58–60). Nominal roots/bases that end in one of a set of specific open syllables (*wa, kwa, pa, da, ra, di, ri, kwi, 'i, ti, chi*) require insertion of a nasal when followed by another morphological element. The examples in (61) show complex nouns that feature the right environment for insertion of the linking nasal (the linking nasal is underlined in the examples below):

(61)
a. maropa-di (papaya-grain, 'papaya seed') → maropa-n-di (H:59)
b. towa-eɬ (path-APPL, 'one's path') → towa:-n-eɬ (H:240)
c. ariwa-maj (top-VLC, 'to be on top') → ariwa-m-maj (H:59)
d. lora-kwa (leaf-ABSL, 'leaf') → lora-[ŋ]-kwa (H:59)

[37] Note that phonological processes that look nonlocal, like vowel harmony, are generally "tier-local".

[38] Although not explicitly noted by H, numerous examples in the work show that nasal assimilation does not happen across word boundaries.

Beyond insertion of the linking nasal, (61) also shows that the linking nasal assimilates in place to an immediately following consonant, and surfaces as [n] elsewhere (e.g., before a vowel). Nasal assimilation is local and purely phonological in Movima.

When the irrealis infix intervenes between an *n* and a following consonant, it bleeds (prevents) the place assimilation of *n*. In (62a), an idiomatic compound places an alveolar nasal before a bilabial, and the nasal predictably assimilates, surfacing as [m]. When the irrealis infix intervenes between the two elements of this compound, (62b), assimilation is bled, and underlying *n* surfaces instead.

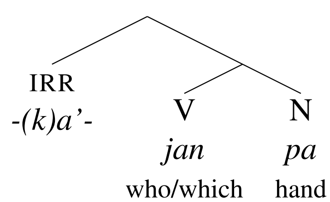
(62) a. jan (who/which) + pa (hand) → jam-pa ‘do’ (H:124)
b. -*(k)a’*- (IRR) + (62a) → jan<*a’*>-pa (H:447)

Similarly, when the irrealis infix intervenes between a linking nasal and a following consonant, the linking nasal’s usual assimilation is bled. In (63a), there is a compound whose form conditions the addition of the linking nasal (underlined), which assimilates to the following *p*; (63b) shows that the linking nasal surfaces as underlying *n* when the irrealis infix intervenes.

(63) a. kweya-m-poy (woman-LINK-animal, ‘female animal’) (H:81)
b. -*(k)a’*- (IRR) + (63a) → kweya-n<*a’*>-poy (H:81)

In both (62b) and (63b), the intrusion of the infix between a nasal and a bilabial consonant in its stem bleeds assimilation of the nasal, such that it remains *n*. Notably, *m* is not an assimilatory consonant (H:34), and so there is no plausible derivation where first *n* assimilates to the following bilabial and becomes *m*, and then unassimilates.

Consider the structure for (62b), where the compound is a constituent and the irrealis scopes over this constituent:

(64)


If nasal assimilation were computed over the constituent containing just the compound, we would expect the *n* of *jan* to assimilate, becoming *m*, prior to the intrusion of the infix. What (62) shows instead is that nasal assimilation is computed only over the whole structure—the word—crucially *including* the irrealis infix. With the intruding infix in its surface (infixed) location, it naturally blocks assimilation.

Another case of word-level phonology being interrupted comes from glottalization: before another consonant (i.e., in coda position), *k* is always realized instead as a glottal (orthographically ’) (H:28). Example (65a) shows glottalization of *k*, underlined, in the absence of an intruding infix.[39] When the irrealis intervenes between *k* and a following consonant, *k* remains *k*, (65b).

[39] Glottalization always applies to word-final /k/, even when the following word begins with a vowel, showing that this is a word-level phenomenon.

(65) a. yok (catch) + -na (DIR) → yo’na ‘catch’ (H:123)
b. *-(k)a’-* (IRR) + (65a) → yok<*a’*>na (H:543)

Importantly, there is no restriction on glottal stop appearing intervocalically, whether root-internally or at a morpheme juncture, so there is no plausible “Duke of York” pathway from *k* to glottal stop (à la (65a)) back to *k* in (65b).

Consistently, then, word-level phonology *is* disrupted by the intrusion of an infix in Movima, in stark contrast to morphophonology in Movima, §3.4. This supports the once-and-for-all computation of word-level phonology. Multistage computation is also possible, so long as only the word-level phonology (and not stem-level or otherwise earlier phonology) gives rise to nasal assimilation and glottalization.

3.5.2 Hunzib: Phonological interactions

A second case study of word-level phonology being disrupted by an infix comes from the verbal plural marker in Hunzib. We saw in §3.4.2 that infixation of the verbal plural marker does *not* disrupt morphophonologically conditioned vowel epenthesis, (56), that is, vowel epenthesis proceeds as though it can’t see the infix. It turns out that the epenthesized vowel, *e*, has even more to tell us. The discussion here again follows Kalin 2022b, which draws on data from van den Berg 1995.

In Hunzib, there is a language-wide word-internal process of vowel harmony, specifically for the vowel *e*: the vowel *e* harmonizes to ə when it follows a central nonlow vowel (i.e., *ɨ* or ə). Compare, for example, (66a) to (66b), both of which feature an epenthesis environment for the causative suffix (see §3.4.2); the vowel undergoing harmony (or not) is underlined.[40]

(66) a. uƛ’ (end) + -k’(e) (CAUS) → uƛ’-k’e ‘make end’ (B:337)
b. ɨx (warm) + -k’(e) (CAUS) → ɨx-k’ə ‘warm up’ (B:308)

When *e* follows an *ɨ* or ə in the preceding syllable, *e* changes to ə, (66b).

When the verbal plural infix intervenes between the root vowel and the harmonizing suffix vowel, the following question arises: will vowel harmony ignore the infix, thereby predicting ə in the suffix when the root vowel is a central nonlow vowel? Or, will vowel harmony include the infix, which crucially does *not* have a central nonlow vowel, thereby predicting the suffix vowel to be *e*? The latter prediction is borne out: vowel harmony does see the infix, so underlying *e* surfaces rather than the harmonized vowel, (67) (B:308).

(67) *-á-* (VPL) + (66b) → ɨx<á>*-k’e* (*ɨx<á>-k’ə)

Infixation bleeds the vowel harmony that would have occurred in its absence, (66b).

[40] Although B makes no specific comment about whether this vowel harmony happens across word boundaries, the texts in the latter half of B show plenty of instances of an *e* in the first syllable of a word, following a word whose final syllable contains *ɨ* or ə, and so I assume that this vowel harmony is a word-bound process.

An anonymous reviewer asks whether this vowel harmony could be phonetic in nature (and thus very “late”), a question I do not have the answer to. However, given that the process is word-bound and reflected in the orthography, my best guess is that it is phonological, and not phonetic.

Given a structure like that in (68), the lack of harmony in (67) points to this phonological process being computed only over the whole word.

(68)

Adj	CAUS	VPL
ɨx	*-k'(e)*	*-á-*
warm		

Vowel harmony must not be computed over just the inner constituent (adjective plus causative), but rather only over a form that includes the verbal plural infix. This supports once-and-for-all computation of the harmony process, or a word-level-specific computation of harmony in a multistage approach.

3.5.3 Overall findings: Phonological interactions

The above case studies are exhaustive in terms of what I was able to identify as a purely phonological process occurring at a morpheme boundary where an infix could intrude. But, there is a variety of further evidence for the sensitivity of regular, word-level phonology to the presence of an infix. In one of the excluded case studies, Turoyo (Kalin 2020), the mobile past tense marker feeds/bleeds all regular phonology, including shortening of geminates, feature-spreading to an empty consonant slot, and vowel lowering. With English expletive infixation, another excluded case study, the surface-level assimilation of *n* to a following velar in a word like *i[ŋ]-competent* is bled by the intrusion of an expletive (*i[n]-<fucking>competent*, **i[ŋ]-<fucking>competent*). Similarly, for some speakers (cf. fn. 35), there is a retreat to underlying *in-* in *im-possible* when the expletive intrudes, giving rise to *in-<fucking>possible*. Further, putting aside cases of intermorphemic infixes, infixes in their canonical intramorphemic positions also feed/bleed regular phonology (Kalin 2022a). An example of this comes from one of the languages in the current sample: in Eton (§3.2), a regular process of high tone spread (Van de Velde 2008:58–60) is bled by the intrusion of an infix that contributes its own tone (Van de Velde 2008:246, (49)). All these cases support once-and-for-all computation of certain phonological processes, at the word level.

In sum, regular, word-level phonology is *not* impervious to the intrusion of an infix, and behaves instead like the infix *is* visible in its intruding position. This type of phonological process must apply only over a domain that already includes the infix, unlike all the other processes/relationships we saw in previous sections. Most straightforwardly, what this supports is once-and-for-all computation of (at least some) phonology.

3.6 Interim summary

We can finally put it all together. Morphophonological, allomorphic, morphosyntactic, and semantic relationships/interactions in the stem of infixation all survive the intrusion of an infix, that is, behave as though the infix isn't there. Word-level phonological interactions do not survive the intrusion of an infix, that is, behave as though the infix is there. What this establishes is a sort of minimal derivational "timeline" (which could be recast as representational levels):

(69) *Step 1:* Within the stem of infixation, establish semantic relationships and morphosyntactic relationships; expone all morphemes; apply morphophonological processes.
Step 2: Add the infix (and any other affixes).
Step 3: Apply word-level phonological processes.

Of course, it would be very strange for only infixation to trigger derivational steps (or separate representational levels), as the minimal "timeline" above seems to suggest. In the following section, I will get more precise about what the architecture of the grammar must look like to naturally give rise to (69). Utilizing the terminology of §2, I will argue in particular that exponence and morphophonology (and some other phonological processes) are computed bottom-up, while (some) phonology is computed once-and-for-all.

4 Implications

When considering phenomena that interface with multiple parts of the grammar, an inevitable big question that arises is *how* the parts of the grammar fit together. In this section, I leverage various aspects of the findings from §3 to argue for a model where (i) exponence and (morpho)phonology are computed bottom-up, with some once-and-for-all phonological computation at the word level (§4.1), and (ii) exponence and (morpho)phonology follow the syntax (§4.2).

4.1 Bottom-up, multistage, or once-and-for-all computation?

In this section, I put forward an outline of the type of model that naturally accounts for all the findings, §4.1.1, and show why other models do not fare as well, §4.1.2. The most crucial evidence throughout revolves around suppletive allomorphy (§3.3), morphophonology (§3.4), and word-level phonology (§3.5); semantic interpretation (§3.1) and morphosyntactic selection/compatibility (§3.2) do not differentiate approaches from each other (as discussed in §2.1).

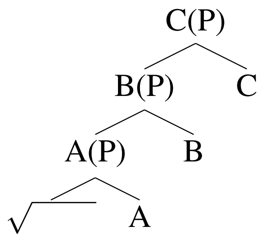
4.1.1 In support of bottom-up exponence and (morpho)phonology, once-and-for-all phonology

The findings in §3 support a mixed model, where exponence and morphologically conditioned phonology are computed bottom-up, while other phonological processes are computed once-and-for-all. This section elaborates on this type of model and how it captures the findings, as well as some other empirical aspects of exponence and infixation. I begin with a discussion about the bottom-up aspects of the supported model before considering once-and-for-all and multistage phonological computation.

As discussed in §2.2.1-2, bottom-up exponence and/or (morpho)phonological processes are features of an otherwise very diverse set of theoretical models (e.g., Kiparsky 1982, Bobaljik 2000, Stump 2001, Inkelas 2008, Embick 2010). In a bottom-up model, the (relevant) operations start from the most embedded component of a morphosyntactic structure (usually a root) and proceed upward. An alternative but similar string-based formulation would hold that operations apply inside-out, assuming that there is some mechanism for determining what the innermost element is, be that through a designation like "root" or "stem" or through a flat representation that encodes embedding in some way.[41] For illustration here, I will assume the structural notion of bottom-up, but the same effects could be accounted for in a string-based, inside-out model.

What exactly is the "bottom" of a structure? Consider the schematic structure in (70), where the terminals are the root and the heads A, B, and C. (I abstract away from the question of whether the nonterminal nodes are phrasal or not, hence the Ps are in parentheses.)

(70)



Building on Myler 2017's calculus for ordering of the operation of exponence, I propose the more general calculus in (71) for determining the domains of application of a bottom-up operation, given a particular structure:[42]

(71) *Bottom-up ordering of an operation*
Given categorially distinct nodes x and y, where:
a. x is the terminal node of a chain of (one or more) projections of x,
b. y is the terminal node of a chain of (one or more) projections of y,
if the highest projection in x's chain dominates y, then (the highest projection of) y undergoes the operation before (the highest projection of) x.

The formulation in (71) establishes the root in (70) as the most embedded node for the purposes of applying an operation bottom-up, followed by A (or its projection A(P)), then B (or its projection B(P)), then C (or its projection C(P)).[43] Bottom-up computation would thus start with the root and proceed step-wise up the structure, terminating at C/C(P).

[41] Note that this is yet distinct from left-to-right computation, based purely on linear order. It is clear from the case studies in the present sample that left-to-right computation is a nonstarter, as there are both left-edge and right-edge infixes in the sample, and there is no differential behavior based on whether an infix is (e.g.) interrupting a prefix and a root versus a suffix and a root.

[42] Myler's definition is given in (i):

(i) Temporal order of Vocabulary Insertion
For a pair of terminal nodes x and y:
If x is the head of a maximal projection M such that M is categorially distinct from y and M dominates y, then y } x.
If y } x, then y undergoes Vocabulary Insertion prior to x.

[43] This formulation does not exhaustively establish ordering over all terminals/projections in all possible structures, just like Myler 2017's formulation. I leave open the question of ordering (or lack thereof) in other configurations apart from the one under consideration here.

How does bottom-up computation account for the results related to exponence and (morpho)phonology? Let's assume for this illustration that we are in a theory like Distributed Morphology, where it is terminal nodes that are subject to exponence. Some toy rules of exponence are given in (72), where A/*-a* and C/*-c* are suffixes and B/*-b-* is an infix:

(72) Rules of exponence

a. $\sqrt{\ } \leftrightarrow$ *root*
b. A $\leftrightarrow$ *-a*
c. B $\leftrightarrow$ *-b-*
d. C $\leftrightarrow$ *-c*

Since we're interested in cases of intermorphemic infixes, let's further imagine that the infixal exponent *-b-* can (incidentally) surface between the root and *-a*, giving rise to the surface linear order *root<b>-a-c*. In the bottom-up exponence of (70), calculated as per (71) utilizing the rules of exponence in (72), there will be four derivational steps (or representational levels), with the string accumulating a new exponent each time a new terminal node is exponed:

(73) Bottom-up exponence

a. root ($\sqrt{\ } \leftrightarrow$ root)
b. root-a (A $\leftrightarrow$ -a)
c. root<b>-a (B $\leftrightarrow$ -b-)
d. root<b>-a-c (C $\leftrightarrow$ -c)

Following Kalin 2022a and the discussion in §2.2.3, I am assuming that infixation of the infixal exponent *-b-* is part of the bottom-up computation. In other words, infixation happens alongside the insertion of *-b-* in (73c); there is no step where *-b-* has been inserted but sits after A/*-a* (**root-a-b*). (See §4.1.2 for a discussion of the undesirable consequences of delaying infixation relative to the insertion of an infix.)

Bottom-up exponence makes the following crucial prediction relevant to the findings of this paper: infixation of *-b-* in between *root* and *-a* should affect neither the choice of exponent for A nor the choice of exponent for the root, because exponence of A and the root, (73a-b), both precede the exponence/infixation of B/*-b-*, (73c). This is a correct prediction—exponent choice in the stem of an infix is unaffected by the intervention of the infix, as seen in the suppletive allomorphy case studies in §3.3. Thus in Nancowry, for example, prosodically conditioned allomorphy of the causative morpheme (corresponding to A in the toy model) is impervious to the intrusion of the outer nominalizing infix (*-b-* in the toy model), as was shown in (41).

There are a variety of other empirical benefits of bottom-up exponence. First of all, when exponent choice is sensitive to phonology/prosody, it is well-established (though not wholly uncontroversial) that the conditioning environment is found *inward* relative to the morpheme being exponed (e.g., Carstairs 1990, Dolbey 1997, Bobaljik 2000, Paster 2006). Second of all, in Kalin 2022a, I report the finding that infixes only ever look/move inward into more (morphosyntactically) embedded material, and never outwardly into less-embedded material. Both empirical observations follow automatically from bottom-up exponence because, at the point of exponence/infixation for a given morpheme, the only visible phonological/prosodic environment is that provided by the morphemes that have already been exponed.

Let's turn now to (morpho)phonology. Bottom-up exponence can be interleaved with bottom-up (morpho)phonology as in (74):[44]

(74) Bottom-up exponence and (morpho)phonology

a. Exponence: *root* ($\sqrt{\ } \leftrightarrow$ root)
b. Cycle of (morpho)phonology over *root*
c. Exponence: *root-a* (A $\leftrightarrow$ -a)
d. Cycle of (morpho)phonology over *root-a*
e. Exponence: *root<b>-a* (B $\leftrightarrow$ -b-)
f. Cycle of (morpho)phonology over *root<b>-a*
g. Exponence: *root<b>-a-c* (C $\leftrightarrow$ -c)
h. Cycle of (morpho)phonology over *root<b>-a-c*

This kind of "radical" interleaving—interspersing exponence and (morpho)phonology with the addition of every exponent—is found in a variety of approaches (e.g., Chomsky and Halle 1968, Kiparsky 1982, Stump 2001, Inkelas 2008, Kalin 2022a). Note that a limited amount of interleaving—at specific stages—is found in multistage approaches as well, which I put aside for now but comment more on below.

Most relevant to the findings of this paper, bottom-up (morpho)phonological computation, along the lines of (74), straightforwardly accounts for the survival of all morphologically conditioned phonology across a linearly disruptive infix, §3.4. In the toy model here, for a hypothetical morphophonological interaction between the root and A/*-a*, the crucial step is (74d); any interaction that's interleaved with exponence will persist even after infixation at a later stage of exponence, (74e). In Movima, (52b) for example, the absolute state suffix triggers morphophonological reduplication that survives intact across the intrusion of the outer irrealis infix.

Although the present language sample uncovered only cases of *morphologically conditioned* phonology surviving across the intrusion of an infix, there is good reason to think that other sorts of phonological processes can be interleaved with exponence as well. First of all, the distinction between "early" and "late" phonology is an old one, encoded in some phonological theories as the cyclic/post-cyclic divide, with cyclic rules able to be purely phonological in nature (Kiparsky 1982, Booij and Rubach 1987, Stump 2001, Bermúdez-Otero 2012, Inkelas 2014). Second, both exponence and infixation show us that at least some prosodification must take place cyclically, from the bottom up: it is not uncommon for infixes to have a prosodic placement (Yu 2007), or for exponent choice to be conditioned by prosodic factors (e.g., Paster 2005). For exponent choice and infixation to be sensitive to regular prosodic processes of a language, it must be that these prosodic processes have already applied before exponence and infixation. Drawing from the sample of case studies in this paper, such cyclic prosodic structure-building is crucial for Movima—where the

[44] Alternatively, bottom-up exponence could properly precede bottom-up (morpho)phonology, though this would require maintaining structure even after exponence has applied (so that (71) could still be utilized), and would make no relevant different predictions for our purposes, so long as infixation patterns with exponence in terms of its timing.

infix is placed after the first foot of its stem[45]—and for Nancowry—where exponent choice for both the causative and the nominalizer is prosodically conditioned.

One might wonder whether the same result could be achieved via multistage computation at specific junctures, along the lines of Stratal OT. There are two reasons to doubt this. First, for multistage computation to work, all (relevant) infixes must sit directly outside an earlier stage/stratum; it should thus be possible in principle to find cases of morphologically conditioned phonology (in the stem of an infix) that does *not* survive the intrusion of an infix; this empirical profile is systematically absent from the present cases studies and the excluded case studies as well.

Second, if phases define strata or the application of the computations above (e.g., Bye and Svenonius 2012, Sande et al. 2020, Rolle 2020), it's important to observe the mix of structures in this sample, not all of which plausibly have a phase boundary between the infix and its stem, as noted a number of times throughout §3. For example, while a phase boundary plausibly exists between category-changing affixes, as in the Nancowry nominalization cases, the same exact effect in Nancowry can be seen between stacked causativizers (§3.3.2). Similarly, in Hunzib, the verbal plural is presumably part of the same phase as other low verbal morphology (§3.4.2, §3.5.2). In the high inflectional domain, a phase boundary plausibly exists between tense/aspect/mood and low verbal morphology (as in the Palauan case study, §3.3.1, and Movima case study, e.g., §3.4.1), but not between agreement and tense/aspect/mood morphology (as in the excluded Turoyo case study, §3.3.3). Phase-based application of exponence and (morpho)phonology is therefore not a panacea to explain the findings without bottom-up computation.

To complete the model, there must be an application of certain phonological processes only over the whole word, once-and-for-all. Recall the findings of §3.5, which show that word-level phonology in the stem of an infix does *not* survive infixation; it therefore must be that some phonological interactions are "late", applying only over larger domains rather than interspersed with exponent choice. This motivates the addition of step (75i) to the toy model:

(75) Bottom-up exponence and (morpho)phonology, once-and-for-all phonology

a. Exponence: *root* ($\sqrt{\ } \leftrightarrow$ root)
b. Cycle of (morpho)phonology over *root*
c. Exponence: *root-a* (A $\leftrightarrow$ -a)
d. Cycle of (morpho)phonology over *root-a*
e. Exponence: *root<b>-a* (B $\leftrightarrow$ -b-)
f. Cycle of (morpho)phonology over *root<b>-a*
g. Exponence: *root<b>-a-c* (C $\leftrightarrow$ -c)
h. Cycle of (morpho)phonology over *root<b>-a-c*
i. Word-level phonology over *root<b>-a-c*

[45] Recall from fn. 10 that penultimate lengthening can feed foot construction and therefore infix placement in Movima, specifically in bisyllabic stems with a light first syllable. Interestingly, the application of penultimate lengthening prior to infixation is apparently sometimes optional, such that the infix has a variable placement with some specific stems (H:82–83; see also fn. 52). Penultimate lengthening is also, independently of this, lexically idiosyncratic to at least a small degree (H:48–49). Given that lengthening can be "undone" when a syllable is no longer penultimate (H:80), I have not found a way to tell whether penultimate lengthening is bottom-up or multistage (e.g., taking place at phase boundaries), but crucially, it is not a once-and-for-all process (at least not obligatorily).

While the processes involved in word-level phonology in this sample could, potentially, be part of a multistage phonological computation, it would be crucial that these processes never are able to occur prior to the stage at which the infix is exponed/infixed (unless they are able to be undone in Duke-of-York fashion). It is therefore most straightforward to simply assume once-and-for-all computation of word-level phonology. The lateness of (some) phonology accounts (for example) for Hunzib, (67). The causative suffix in Hunzib undergoes both morphophonological vowel epenthesis (§3.4.2) and purely phonological vowel harmony (§3.5.2). Vowel epenthesis takes place bottom-up, and therefore survives the intrusion of the verbal plural infix. Harmony of the epenthesized vowel takes place as part of once-and-for-all phonology (at step (75i) above), and so does not survive across the infix.

4.1.2 Against once-and-for-all exponence and morphophonology

As discussed in §2.2, once-and-for-all computation is popular within a number of OT-based models (e.g., McCarthy and Prince 1993a,b, Prince and Smolensky 1993, Mester 1994, Kager 1996, Mascaró 1996, Benua 1997, Bonet et al. 2007, Steriade 2008, Zukoff 2023) and other constraint-based models (e.g., Bonami and Crysmann 2016, Crysmann and Bonami 2016). In these models, well-formedness of (morpho)phonology and (some types of) exponent choice is evaluated simultaneously, over an entire word. As discussed in §2.2.3, such models also generally incorporate infixation as part of the once-and-for-all computation (e.g., Hyman and Inkelas 1997).

To illustrate, consider Mascaró 2007's analysis of Djabugay (originating in an earlier version in Kager 1996, who in turn cites Patz 1991), a Pama-Nyungan language of Australia. In Djabugay there is suppletive allomorphy of the genitive morpheme, as *-n* after a vowel and *-ŋun* after a consonant. For Mascaró 2007, this choice is determined in the phonology, through an interaction of the following constraints: PRIORITY, "which enforces priority of dominant morphs", MAX and DEP, which penalize the addition or deletion of input material, and COMPLEX, which penalizes complex codas. The input to the phonology has both potential exponents for the genitive, with priority assigned to *-n*. Two relevant computations are shown in (76), where subscripted numerals index specific exponents, and > indicates priority (Mascaró 2007:729).

(76) Mascaró 2007's once-and-for-all approach to exponence

a.

guludu-{n_1>$ŋun_2$}	COMPLEX	MAX, DEP	PRIORITY
a. ☞ gu.lu.du-n_1			
b. gu.lu.du.-$ŋun_2$			∗!

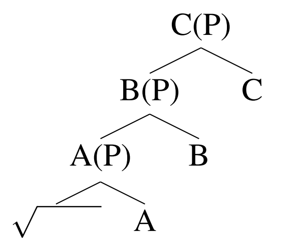
b.

gaŋal-{n_1>$ŋun_2$}	COMPLEX	MAX, DEP	PRIORITY
a. ga.ŋal-n_1	∗!		
b. ga.ŋal.-n_1a		∗!	
c. ☞ ga.ŋal.-$ŋun_2$			∗

For vowel-final stems, as in (76a), the prioritized exponent *-n* wins. For consonant-final stems, as in (76b), the prioritized exponent is not chosen because it creates a violation of COMPLEX;

this violation is resolvable without violating the MAX and DEP constraints through the choice of the nonprioritized exponent *-ŋun*. The crucial point is that the choice among exponents is being regulated in the phonological grammar, alongside phonological constraints. The same is true of once-and-for-all approaches to (morpho)phonology and infixation.

Returning to the schematic structure in (70), repeated below, once-and-for-all computation of (morpho)phonology, exponence, and infixation would take these all to be determined simultaneously in the phonological grammar.

(77)


Assuming the same exponents as given in §4.1.1 for this toy example, candidate strings would include both *root-a-b-c* (without infixation of *-b-*) and *root<b>-a-c* (with infixation of *-b-*). Importantly, given that these are competing candidates and not steps in a derivation, the winning/attested ordering—*root<b>-a-c*—leaves no room for adjacency of the two morphemes the infix intrudes between. Candidates would also include all possible combinations of suppletive (competing) allomorphs with (non)infixation of *-b-*.

There are a number of undesirable predictions of such a model with respect to interactions between an infix and its stem. First, suppletive allomorphy arising in the interaction between two morphemes (e.g., the root and *-a* in the toy example) should be disrupted by the infix (*-b-*), because in the string over which appropriateness of different allomorphs would be evaluated, *root<b>-a-c*, the infix is already in an infixed (disrupting) position. Second, two morphemes split up by the infix should not interact (morpho)phonologically in any capacity, again because the one available string over which such interactions could apply, *root<b>-a-c*, already has *root* and A/*-a* nonlocal to each other.[46] Both of these predictions are, of course, incorrect, as shown in §3.3.3 and §3.3.4.

Another argument against once-and-for-all computation comes from earlier work of mine on infixes (Kalin 2022a). From a survey of 32 morphemes that exhibit suppletive allomorphy, where at least one allomorph is infixal, I find that such allomorphy is always determined from the stem edge, and never from the surface position of an infix. This is very surprising if exponence and infixation take place once-and-for-all, again since *root<b>-a-c*—with the infix in its surface location—is the only string available for evaluating suppletive allomorph choice. Also noted in Kalin 2022a is that, in a once-and-for-all model, there is no a priori reason why an infix wouldn't be able to displace into (or even see) less embedded phonological material, since that material is available simultaneous with inner material. Finally, in Kalin 2022a I show that infix placement can be opaque, with the location of the infix needing to be stated with respect to a pivot that either disappears or is nonadjacent to the infix on the surface, due to (later) phonological processes. All of these findings are naturally dealt with in a bottom-up model of exponence and (morpho)phonology (§4.1.1).

[46] An exception, of course, would be if the suppletive or (morpho)phonological process at hand is one that happens exceptionally at-a-distance, which should be evident from other environments.

I will explore two possible ways to "save" once-and-for-all computation of exponence and/or (morpho)phonology. First, it has long been known that there are cyclic/derivational effects in the grammar, and correspondingly, there are OT-based mechanisms that have been proposed to account for them. Some of these essentially reproduce derivational stages, including Optimal Interleaving (e.g. Wolf 2008, based on McCarthy 2007) and Harmonic Serialism (e.g, McCarthy 2000, Müller 2021, Gleim et al. 2023), and so such solutions are indeed able to account for the findings here, as (I would argue that) they are at their core bottom-up approaches. The other prominent OT-based solution to cyclic effects, which aims to maintain once-and-for-all computation, is Output-Output correspondence theory (originating with Kenstowicz 1996, Benua 1997, Burzio 1998; see the typology of such theories in Rolle 2018). Essentially, what this adds to the phonological grammar is the possibility of evaluating candidates for faithfulness to a related word form.

To capture the cyclic effects from §3 using Output-Output correspondence theory, the natural comparison word form would be one lacking an infix. For example, candidates for (78a) would be compared with (78b).

(78) a. root<b>-a-c
b. root-a-c

A local effect—e.g., a (morpho)phonological or allomorphic interaction between the root and -*a*—could then appear in the word *with* the infix, (78a), if the interaction exists locally in a related word *without* the infix, (78b). However, a significant problem arises: in once-and-for-all computation, all of the relevant operations—exponence, morphologically conditioned phonology, and word-level phonology—are on par. But, we know from §3 that only the former two types of computations are faithful to the related word form, while word-level phonology would necessarily *not* be. This difference could be stipulated, via specific Output-Output constraints. But the systematicity of the results reported here, in terms of the split between exponence and morphophonology on the one hand, and phonology on the other, would be completely accidental, with the reverse sorts of patterns expected to be possible as well. In addition, Output-Output correspondence theory cannot straightforwardly account for the other issues with once-and-for-all computation noted above (emerging from Kalin 2022a), in particular, the stem edge as the location of exponent choice, the inward directionality of infixation, and the potential opacity of infix placement. I therefore reject this type of account of the present findings.

A second, different way to salvage once-and-for-all computation would be to take the operation of infixation—and just this operation—to be delayed relative to exponence and (morpho)phonology. In other words, there would be an intermediate stage *root-a-b-c*, (79a), separate from the string with the infix in its surface position, *root<b>-a-c*, (79b).

(79) a. root-a-b-c (once-and-for-all computation)
b. root<b>-a-c (delayed infixation)

It's crucial to note that the pre-infixation string in (79a)—which maps the morphosyntactic ordering to a transparent/faithful linear order—is a string that is not generated under any other model considered in this or the previous section. It is precisely this string, schematized in (79a), that will both sidestep some of the empirical problems with full simultaneity. But, it will also create some new problems, detailed below.

At a first glance, a model that incorporates once-and-for-all exponence and (morpho)phonology with delayed infixation fares better than the truly once-and-for-all model. Since there is a string—(79a), *root-a-b-c*—where there is adjacency between the root and A/-*a*, these morphological elements can interact in terms of suppletive allomorphy and (morpho)phonology, without the intrusion of the infix between them. This is, of course, a desirable prediction (§3.3–4).

However, applying exponence and (morpho)phonology based on the string in (79a) makes a number of *un*desirable predictions as well. First of all, word-level phonology should apply over that same string (unless it is further delayed until after infixation) and thus survive the later intrusion of the infix, counter to fact (§3.5). Second, -*b*- in (79a) should be able to trigger and undergo (morpho)phonological processes, in its nonsurface location. As discussed in both Yu 2007 and Kalin 2022a, infixes do not look like they live any kind of exponed/phonological life in their underlying (noninfixed) position—infixes are *only* visible for conditioning and undergoing (morpho)phonology in their infixed (surface) positions. Third, the string in (79a) predicts that suppletive allomorphy of C could be phonologically conditioned by the adjacent -*b*-, when in fact relevant case studies show that infixes must be in their infixed position already for the purposes of phonologically conditioned suppletive allomorphy of further-out morphemes (Kalin 2022a, 2023). And some other issues for once-and-for-all computation persist for this model, including the inward directionality of infixation, the inward directionality of phonologically conditioned suppletive allomorphy, and the potential opacity of infix placement.[47]

I therefore conclude that a once-and-for-all model fails to capture the data at hand, and faces additional problems with respect to infixes and infixation.[48]

4.2 In support of post-syntactic exponence and (morpho)phonology

This section turns to the relationship between morphological and phonological computation on the one hand to syntactic computation on the other hand. Bottom-up exponence and (morpho)phonological computation are features of an otherwise very diverse set of theoretical models, varying from each

[47] An anonymous reviewer suggests a hybrid model that could save once-and-for-all exponence. The model would work as follows. Exponence would be computed once-and-for-all over a morphosyntactic structure, thereby inserting the exponents into the structure. (Morpho)phonology would then proceed from the bottom of this structure upward, hyper-cyclically. Infixation would take place during this bottom-up (morpho)phonological computation, at the point where the infixal exponent is incorporated with its stem. This model would correctly predict that an infix would neither disrupt exponence of further-embedded morphemes nor (morpho)phonological interactions among them.

However, this hypothetical model would not be able to capture basic facts about exponence and so is not a viable model. Specifically, it is well-established that exponence of a morpheme can be sensitive (inwardly) to its phonological/prosodic environment (e.g., Carstairs 1987, 1988, 1990). If insertion of exponents is simultaneous across a structure, then exponent choice should not be able to be sensitive to the phonological forms of other exponents, inwards or outwards. It is for this reason that once-and-for-all exponence, in all actual models that posit this mode of exponence, is always computed via a competition in a constraint-based grammar that compares multiple possible outputs. In addition, exponent choice is able to be conditioned by amalgamated prosodic structure in its stem (e.g., Paster 2005), meaning that amalgamation and at least prosodification can't be wholly delayed until after all of exponence.

[48] While I did not consider in detail the possibility that realization could be top-down in addition to bottom-up, all evidence from infixation points to a uniform bottom-up model, including the data presented in this paper. See Kalin 2022a for a discussion of bidirectional models.

other in particular in the relation of these types of computations to syntax. Bottom-up exponence and/or (morpho)phonological computation might take place in a grammatical module that (i) is prior to the syntactic computation, generating the input to the syntax (e.g., Kiparsky 1982, Wunderlich 1996, Müller 2021), (ii) runs separate from but in parallel to the syntactic computation (e.g., Paradigm Function Morphology; Stump 2001, 2016), or (iii) is after (fed by) the syntactic computation (see in particular specific versions of Distributed Morphology; Bobaljik 2000, Embick 2010, Myler 2017).

How does the behavior of infixes at morpheme boundaries, as explored at length in §3, bear on the question of where morphology and phonology fit into the grammar? While the literature already offers a variety of evidence for a post-syntactic, "late insertion" view of exponence and (morpho)phonology (see, e.g., the arguments compiled in Kalin and Weisser To appear), the data presented in this paper furnish a novel argument for lateness. Specifically, this evidence comes from the diversity of morphosyntactic units into which infixes can lodge themselves. The basic arc of the argument is as follows: given that infixation is an aspect of exponence and (morpho)phonology, anything that feeds the infixation process itself must precede these computations; there are syntactic operations that feed infixation; therefore, syntax precedes exponence and (morpho)phonology.

The relevant case at hand here is object incorporation in Movima. Recall from §3.2.1 that the irrealis infix in Movima—which appears after the first foot—can infix into object incorporation structures, as seen in (80b) (example repeated from (27)):

(80) a. iń jiɨ-a:-pa
1ABS grate-DIR-manioc
'I grate manioc.'
b. iń jiɨ-a<*ka'*>-pa
1ABS <IRR>grate-DIR-manioc
'I'll grate manioc.' (H:79)

There is a variety of evidence suggesting that the process of object incorporation itself is syntactic in Movima (*à la* Baker 1996). First of all, it is only an underlying patient that can be incorporated into the verb (Haude 2019:243). Second, notice the behavior of the direct voice marker, *-a*, "whose basic function is to indicate that the verb is semantically bivalent and that its [absolutive argument] is the undergoer" (H:323).[49] This marker is obligatory on the verb even when the patient is incorporated and the agent is marked absolutive, as in (80). Finally, the incorporation structure does not look like compounds do in Movima, which are generally right-headed and nominal in nature (H:§5.2.1). Nevertheless, incorporated objects behave (morpho)phonologically like they are fully

[49]The voice system and general clause structure of Movima are complex (Haude 2019). Direct voice is used when the agent argument outranks the patient argument in terms of person, animacy, and topicality; in this voice, the agent is adjacent to the predicate and forms a prosodic constituent with it. Inverse voice is used when instead the patient outranks the agent, and in this case, the patient is adjacent to the predicate and forms a prosodic constituent with it. Intransitive predicates generally appear without voice marking, and the sole argument of the intransitive patterns with the patient in direct voice (an absolutive pattern) and the agent in inverse voice (a nominative pattern). Object incorporation is restricted to the patient in direct voice, and not possible for any argument in inverse voice. I do not attempt to fully explain this patterning here, but believe it is compatible with object incorporation being syntactic.

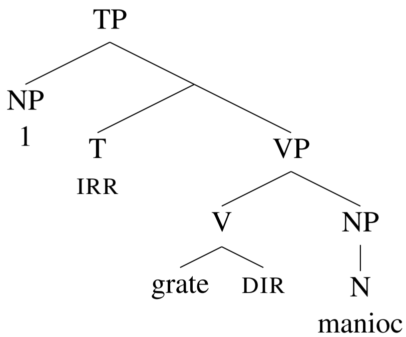
integrated into the word domain in Movima, including in the positioning and behavior of the infix.

If object incorporation is syntactic, and the stem of the infix in (80b) is the verb plus the incorporated object, then it follows that infixation must be *post*-syntactic. This derivation is shown in (81). (81a) provides a bare-bones schematized version of object incorporation as head movement, followed by head movement of V to T. (81b) shows exponence/infixation happening post-syntactically. (Note that the thrust of the argument is compatible with different ways to incorporate the object and build the verb word. Note also that, as per late-insertion models, the syntactic terminals in (81) have lexical/functional identities but no phonological forms.)

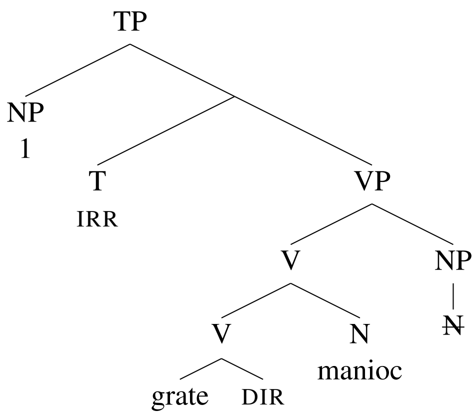
(81) A post-syntactic account of infixation into object incorporation structures

a. Syntax (object incorporation):

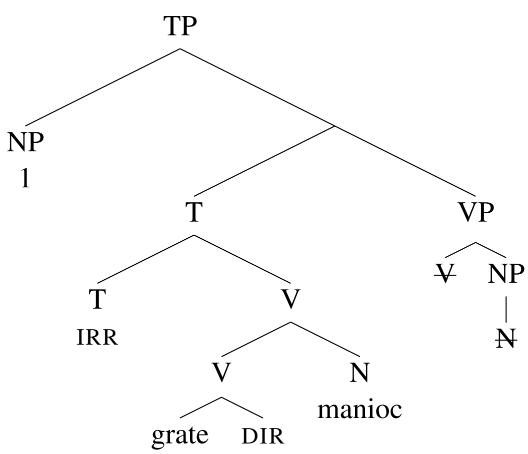
(i)



(ii)



(iii)



b. Post-syntactic morphology (partial snapshot):
   (i) Bottom-up exponence up to the highest V node: jiɨ-a-pa[50]
   (ii) Exponence/infixation of T (next-higher node): jiɨ-a<*ka’*>-pa

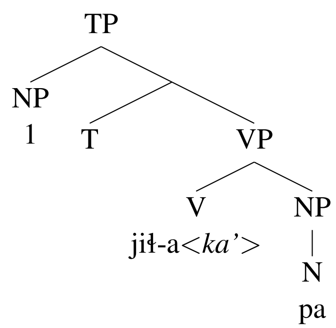
The syntax generates the full verb-word structure, shown in (81a), followed by bottom-up exponence of the whole word, (81b). (I leave out (morpho)phonology for now.)

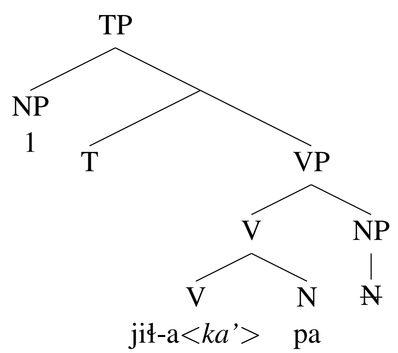
A careful reader might notice that the position of the infix in (80b) is, hypothetically at least, compatible with infixing the irrealis marker into a pre-syntactically assembled stem that consists *just* of the verb and direct marker, (82a), and then in the syntax incorporating the object at the end of this stem, (82b):

(82) A pre-syntactic account of infixation into object incorporation structures

a. Pre-syntactic morphology (partial snapshot):
   (i) Exponence of verb word (incl. infixation): jiɨ-a<*ka’*>

b. Syntax (object incorporation):

(i)


(ii)


Such an analysis would open the door for infixation to precede the syntax, as in pre-syntactic approaches to morphology (e.g., Wunderlich 1996, Müller 2021), or to take place in parallel to the syntax (e.g. as in Paradigm Function Morphology). Importantly, in these types of approaches, morphophonology (and some other kinds of phonology as well) also apply in the morphological module (pre-syntactically or parallel to syntax), while other phonology applies post-syntactically.

[50]Recall from fn. 15 that *pa* itself is a bound form of the root meaning ‘manioc’. This is naturally accounted for under a post-syntactic approach to exponence since *pa* is already incorporated into the verb word at the point of the exponence of the root.

A quirk of Movima morphophonology rules out the possibility of a derivation like (82), and shows that infixation really must happen *after* object incorporation. The illuminating process works as follows: whenever the irrealis infix incidentally lands word-finally (i.e., when the stem in its entirety is a foot), this triggers infixing CV reduplication before the final syllable of the word (H:82). Consider the following examples. When the irrealis marker combines with the verb *joy* ('go') or the noun *michi* ('cat') bearing no other affixes, the infix—in taking its usual placement after its pivot, the first foot—will appear at the end of the word (*joy<a'>*, *michi<ka'>*). Since the infix is word-final in this infixed position, this triggers an exceptional morphophonological process where the initial CV of the final syllable of the resultant verb form (*ya* for *joya'* and *ka* for *michika'*) is reduplicated; this CV reduplicant is then itself infixed before the final syllable. These pieces come together in (83a–b), where the final form features a verb-final irrealis marker and an infixed CV reduplicant.[51] (In the examples below, the irrealis marker is italicized, and the reduplicant is underlined; note that vowel length in the reduplicant is the result of penultimate lengthening.)

(83) a. -*(k)a'*- (IRR) + joy (go) → jo<ya:>y<*a'*> (H:274)
b. -*(k)a'*- (IRR) + michi (cat) → michi<ka:><*ka'*>[52] (H:83)

Infixing reduplication is clearly morphologically conditioned phonology as it occurs specifically for the irrealis morpheme, given a very particular phonological environment, and is not a general property of the phonology of Movima. It is also an unmistakeable indicator of cases where the irrealis infix lands word-finally.

Let's now turn back to the object incorporation cases and how this morphophonological process sheds light on derivational timing (or representational levels). If infixation is post-syntactic, that is, takes place after object incorporation, (81), then the stem of the irrealis infix will include the incorporated object, such that *-(k)a'-* has *jiɨ-a-pa* as its stem, (81b). Given a stem of this size, no infixing reduplication is predicted, because the irrealis marker is never word-final. If, on the other hand, infixation were pre-syntactic, that is, were to take place before object incorporation, (82), then the stem of the irrealis infix would *not* include the incorporated object, such that *-(k)a'-* has just *jiɨ-a* as its stem, (82a). The morphological module thus produces *jiɨ-a<ka'>*, with the irrealis marker in a word-final position. Importantly, since infixing reduplication is clearly morphophonological in nature, we would expect this process to also be triggered in the morphological module, and to therefore be triggered at this stage, incorrectly generating *jiɨ-a<ka><ka'>*. Since there is no infixing reduplication in object incorporation cases, it must be that the infix is not word-final at any relevant derivational point or level of representation. Putting it all together: exponence, infixation, and (morpho)phonology must follow the syntax.

[51] I make no claims here about the relative ordering between reduplication and infixation of the reduplicant, as the data are compatible with either ordering—reduplicate, then infix the reduplicant before the final syllable, or, infix the (empty) reduplicant before the final syllable, then reduplicate.

[52] H:83 notes that *michi<ka:><ka'>* alternates with *mi<ka'>chi*. The latter form is straightforwardly understandable as resulting from the variable timing of penultimate lengthening noted in fn. 45. If penultimate lengthening precedes infixation, then the first syllable will be long and constitute a foot, such that the infix is placed word-internally rather than word-finally.

5 Conclusion

In this paper, I have reported novel cross-linguistic findings related to the (in)visibility of infixes when they appear incidentally at a morpheme juncture in their stem. The relationships and interactions that *see through* an intruding infix must be established *prior to* infixation (or stem from prior relationships/interactions). The relationships and interactions that *do not see through* an intruding infix must be established *after* infixation. The data therefore afford us a novel way to probe at interactions among phonology, morphology, syntax, and semantics.

I have argued that the findings provide evidence for the following specific architectural aspects of morphology and phonology: (i) morphology is post-syntactic (à la Halle and Marantz 1993, 1994); (ii) exponent choice proceeds from the bottom of a structure upward (Bobaljik 2000, Embick 2010, Myler 2017, Kalin 2020, i.a.); and (iii) bottom-up exponence is interleaved with some (morpho)phonological processes, but not others (Kiparsky 1982, Booij and Rubach 1987, Stump 2001, Bermúdez-Otero 2012, Inkelas 2014, Kalin 2022a, i.a.). Other findings in this paper relate to the need to recognize a derivational step or level of representation that establishes semantic relationships (§3.1) and morphosyntactic relationships (§3.2) separately from the phonological location of an infixal exponent; this is a possibility afforded across all generative theoretical models to my knowledge.

Of all of the findings, I take bottom-up exponence and (morpho)phonology to be the most central and robust. Indeed, there is converging evidence for such bottom-up computation from a variety of domains, including suppletive allomorphy (Carstairs 1987, 1990, Dolbey 1997, Paster 2006, Embick 2010), infixation more generally (Kalin 2022a), replacive grammatical tone (Rolle 2018), nonlocal phonological interactions in Mirror-Principle violating structures (Myler 2017), and portmanteau formation (Banerjee 2021). Despite this converging evidence, it is hard to rule out the possibility of multistage computation, given the diversity of such approaches and their inherent flexibility. But, what I hope to have shown is that given two extreme possibilities (bottom-up computation and once-and-for-all computation) and a broad spectrum in between (multistage computation), the facts push us closer to the bottom-up side of the spectrum than the once-and-for-all side. There are also of course exceptional cases that seem to break the bottom-up generalizations, and I leave it as an open question whether all such cases can be reanalyzed as due to (e.g.) exceptional morphosyntax (e.g., Kalin 2020) or exceptional phonology (e.g., Kiparsky 2021).

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